

The Prime-Spectral Framework: A Rigorous Reconstruction

The mathematics of the primon gas elevated to theorem grade: the self-adjoint primon Hamiltonian with simple spectrum $\hbar\omega_0 \log n$, the partition function $\zeta(\beta\hbar\omega_0)$, the Hagedorn temperature, the exact counting law, the Page–Wootters zeta clock, and the repaired arithmetic entanglement—bridging the gaps of the original framework and demoting every unproven claim to a falsifiable twelve-entry conjecture register.

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Abstract

We develop the mathematical and physical content of the “prime-spectral framework”: the statistical mechanics of a free Bose gas of “primons” (one-particle energies $\log p$, p ranging over the rational primes), advanced as the arithmetic core of a unified picture of quantum gravity, time, and the arrows of time. The mathematical core is theorem-grade. The primon Hilbert space is ℓ^2 of the free commutative monoid on the primes, which sidesteps the nonseparability of naive infinite tensor products; the primon Hamiltonian is proved self-adjoint with simple spectrum $\hbar\omega_0 \log n$; the partition function is the Riemann zeta function, whose pole at $\beta\hbar\omega_0 = 1$ fixes a Hagedorn temperature $T_H = \hbar\omega_0/k_B$ and confines the free Gibbs state to $T < T_H$; the state-counting law $N(E) = \lfloor e^{E/\hbar\omega_0} \rfloor$ holds with a defect that is exponentially small in E ; and the statistics of composite states follow the Hardy–Ramanujan normal-order and Erdős–Kac theorems. Three obstructions to the framework’s dynamical claims are then discharged: the diagonal primon Hamiltonian admits no transitions, so an interaction sector (multiplication operators and hopping dynamics on the *prime lattice*) is introduced; composite integers are *not* entangled states (the occupation basis is a product basis), so arithmetic entanglement is defined across prime-mode bipartitions; and exact periodicity is forbidden by the $\mathbb{Q}$ -linear independence of prime logarithms while approximate recurrence survives in every finite truncation, with return times bounded polynomially in the entropy. Claims whose proof is out of reach are consolidated into an explicit, falsifiable conjecture register C1–C12, from the zeta/black-hole partition-function identification to a Galois–Langlands dictionary for the Standard Model. Two register entries are tested numerically at laptop scale: eigenstate-thermalization diagnostics for C4 on truncations up to $D \approx 6 \times 10^4$ (eigenwindows), $D \approx 10^5$ (Krylov trajectories), and, past the *LU*-fill ceiling of the interior eigensolver, $D = 3.8 \times 10^4$ by a factorization-free Chebyshev tier, against label-scrambled controls with the same graph and density of states—the kinetic completion relaxes to its diagonal ensemble, while its eigenstate fluctuations decay as $D^{-0.19}$ ($d = 3$) and $D^{-0.22} \rightarrow D^{-0.26}$ ($d = 4$, extended past the ceiling), receding from the strong-ETH scaling $D^{-1/2}$ —and a pre-registered Chebotarev-dictionary test for C9, which excludes the signature at the Z pole with unit detection power over the registered range of group orders. The mathematical core of the C4 question is reduced to occupation sectors that are exactly block-tridiagonal in a layer grading, solvable in closed form at two modes, and governed by a single interaction whose classical sector limits split integrable from chaotic; and the dictionary instrument runs table-agnostic on frozen extracts, excluding at the W scale and flagging a precision-limited anomaly at the Higgs scale. The result is a framework whose theorems are proved or cited exactly, whose analogies are tightened to citable results, and whose remaining ambitions are stated with enough precision to be confirmed or refuted.

1 Introduction

1.1 The framework

The prime-spectral framework proposes that the prime numbers, through the primon gas of Julia [1] and Spector [2], supply the quantum-mechanical spectrum of the near-singularity gravitational field, that the Riemann zeta function is the gravitational partition function, and that the architecture of time—its problem, its arrow, its block-universe metaphysics, and its impossibility of cycles—follows from the arithmetic of the integers. Recent developments linking conformal primon gases to black-hole interiors [3], the philosophical literature of time’s arrow [4–6], and first-person metaphysics [7–9] motivate the construction. Its ambitions extend to Gaussian primes as a five-dimensional chirality-carrying sector, the Absolute Galois group as a grand unified symmetry, and gravity as an entropic, $1/N$ -suppressed collective effect.

The framework alternates between genuine mathematics, physics speculation, and rhetoric; the skeleton is unusually coherent, while the joints are loose. This paper keeps the skeleton, tightens every joint, and labels everything else. Three instruments suffice: state precisely what can be proved; construct what is missing; and demote what remains to falsifiable conjectures.

1.2 Epistemic status: prove, construct, demote

Every substantive statement in this paper receives one of three labels.

1. **Prove.** A statement with a precise mathematical formulation and a proof is stated as a theorem, lemma, or proposition with its proof or an exact citation. Examples: the $\mathbb{Q}$ -linear independence of $\{\log p\}$; the identification of the partition function with ζ ; the Hagedorn asymptotics; the positive-definiteness of the Page–Wootters clock kernel; the exact state-counting identity $\log N(E) = E/\hbar\omega_0 + O(1)$.
2. **Construct.** Where a mechanism is missing, it is supplied and its consequences proved. Three constructions dominate this paper: (i) the diagonal primon Hamiltonian cannot drive a “cascade into composite states”, so a multiplicative transition sector (multiplication operators M_n and a hopping dynamics on the *prime lattice*) is introduced in Section 3; (ii) composite integers are *not* entangled states—the occupation basis is a product basis—so arithmetic entanglement is defined across prime-mode bipartitions in Section 7, and “arithmetic holism” is relocated from single integers to superpositions; (iii) the “universe cannot repeat” claim conflates exact periodicity, Poincaré recurrence, and dynamical return; these are separated in Section 4 into one theorem and two counterexamples.
3. **Demote.** A claim that is plausible, well-formed, and beyond current proof technology is restated as a numbered conjecture with evidence, dependencies, and a falsifier. Twelve conjectures (C1–C12) are consolidated in Section 10. The guiding rule is that a conjecture must be *operationally testable*: each of C1–C11 carries at least one specified route to confirmation or refutation. Interpretive entries (grade C) are exempt from this rule and marked as such: they are tethered reconstructions, not standalone empirical claims.

Numerical evidence is held to the same standard: every diagnostic is defined before it is computed, and finite-size results are reported with their uncertainty and their dependence on protocol choices (window placement, pivoting, trajectory horizon).

1.3 Summary of results

The framework acquires a layered structure. The *mathematical floor* (all theorem-grade) consists of: the arithmetic Hilbert space $\mathcal{H}_A = \ell^2(\mathbb{N}_{\geq 1}, \times)$ (Section 2); the self-adjoint primon Hamiltonian with simple spectrum $\hbar\omega_0 \log n$; the partition function $Z(\beta) = \zeta(\beta\hbar\omega_0)$ with Hagedorn temperature $T_H = \hbar\omega_0/k_B$; the counting law $N(E) = \lfloor e^{E/\hbar\omega_0} \rfloor$ with exponentially small defect; the Erdős–Kac statistics of the microcanonical ensemble; the isometric representation $n \mapsto M_n$ of the multiplicative monoid on $\mathcal{H}_A$; the phase-incommensurability theorem forbidding exact periodicity, with quantitative return-time bounds; the Bohr almost-periodicity of the zeta clock’s correlation kernel; and the product-basis theorem that fixes where arithmetic entanglement lives. The *conjectural superstructure* consists of the twelve register entries, of which the deepest are C1 (the zeta/black-hole identification), C3 (the arithmetic holographic bound), C4 (thermalization on the prime lattice), and C9 (the Galois–gauge dictionary). Two entries carry carried-out numerical programs: eigenwindow and Krylov diagnostics for C4 to $D \approx 6 \times 10^4$ and $D \approx 10^5$ respectively, with a factorization-free Chebyshev tier to $D = 3.8 \times 10^4$ and an upper-edge Krylov window to $D = 1.05 \times 10^5$ (Remark 3.12), and a pre-registered dictionary application for C9 that excludes the Z -pole branching table with unit detection power over group orders $N \leq 360$.

at current precision, confirmed at seven-channel granularity against a frozen 2024 PDG extract (Section 11).

The philosophical theses of the framework—eternalism with perspectival observers, an extrinsic arrow grounded in arithmetic state-counting, derivative causality, and the rejection of superdeterminism—survive in *conditional* form: they are well-defined consequences of the conjectural superstructure, and they are marked as interpretive rather than theorem-grade wherever the mathematics cannot carry them (Section 9).

2 The Arithmetic Sector

This section fixes the foundational objects. Everything here is elementary but must be stated carefully, because a naive formulation writes the primon Fock space as an infinite tensor product, an object that is *not* separable and is the wrong home for the theory. The construction is cheap and clarifying. *Units and conventions.* Throughout, $k_B = 1$ unless it is explicitly displayed; inverse temperature β is measured in inverse energy, so that $\sigma = \beta \hbar \omega_0$ is dimensionless. The clock regulator of Section 5 is a separate dimensionless parameter, denoted ν there to avoid a collision of symbols.

2.1 The arithmetic Hilbert space

Let $\text{Pr} = \{2, 3, 5, \dots\}$ denote the rational primes. Let

$$\mathcal{F} = \left\{ k: \text{Pr} \rightarrow \mathbb{N} : k_p \neq 0 \text{ for only finitely many } p \right\} \quad (1)$$

be the free commutative monoid on the primes. By the fundamental theorem of arithmetic, the map $\mathcal{F} \rightarrow \mathbb{N}_{\geq 1}$, $k \mapsto \prod_p p^{k_p}$, is a monoid isomorphism between $(\mathcal{F}, +)$ and $(\mathbb{N}_{\geq 1}, \times)$, where $+$ denotes componentwise addition of occupation vectors. This single classical theorem is the load-bearing wall of the entire framework: it is what makes “states of the arithmetic universe” and “positive integers” the same bookkeeping.

Definition 2.1 (Arithmetic Hilbert space). The *arithmetic Hilbert space* is

$$\mathcal{H}_{\mathcal{A}} = \ell^2(\mathcal{F}) \cong \ell^2(\mathbb{N}_{\geq 1}), \quad (2)$$

with orthonormal basis $\{|k\rangle\}_{k \in \mathcal{F}}$, written $|n\rangle$ for $n = \prod_p p^{k_p}$ when no confusion arises. Each prime p defines a *mode* with occupation counter

$$\hat{n}_p |k\rangle = k_p |k\rangle, \quad (3)$$

a self-adjoint operator with spectrum $\mathbb{N}$.

Remark 2.2 (Why not an infinite tensor product?). A naive formulation writes $\mathcal{F}_{\text{Pr}} = \bigotimes_{p \in \text{Pr}} \mathcal{F}_p$ with one bosonic oscillator per prime. The literal von Neumann infinite tensor product of the $\ell^2(\mathbb{N})$ over a countably infinite index set is nonseparable, and its physically relevant separable subspaces (the incomplete tensor products) each require a choice of reference vector. The choice is forced here: the multiplicative theory lives in the incomplete tensor product over the vacuum $\Omega = \bigotimes_p |0\rangle_p$, and Definition 2.1 achieves it directly: ℓ^2 of the countable basis $\mathcal{F}$ is separable, the unitary $|k\rangle \mapsto \bigotimes_p |k_p\rangle_p$ identifies it with both von Neumann’s incomplete tensor product with reference Ω and the bosonic Fock space $\Gamma(\ell^2(\text{Pr}))$, and the tensor factorization over any *finite* set of modes is available whenever needed (Proposition 7.1). The nonseparable full product is never used. This is the first, and easiest, instance of the general principle of this paper: informal infinite constructions are replaced by the countable objects they were meant to describe.

For each prime p the standard bosonic ladder operators act on the p -mode: on basis vectors

$$a_p^\dagger |k\rangle = \sqrt{k_p + 1} |k + e_p\rangle, \quad a_p |k\rangle = \sqrt{k_p} |k - e_p\rangle, \quad (4)$$

where e_p is the unit occupation vector, and $[a_p, a_q^\dagger] = \delta_{pq}$ holds on the dense span of basis vectors. The total particle number is $\hat{N}_{\text{tot}} = \sum_p \hat{n}_p$, self-adjoint on $\{\psi : \sum_p k_p^2 |\psi_k|^2 < \infty\}$.

2.2 The multiplicative number operator and the primon Hamiltonian

The central operator of the framework is the *multiplicative number operator*, whose eigenvalue on $|k\rangle$ is the integer $n = \prod_p p^{k_p}$ itself. We define it together with the Hamiltonian it generates.

Definition 2.3 (Multiplicative number operator; primon Hamiltonian). The *multiplicative number operator* $\hat{N}$ is the diagonal operator

$$\hat{N}|k\rangle = \left(\prod_p p^{k_p}\right)|k\rangle, \quad (5)$$

self-adjoint on its maximal domain. The *primon Hamiltonian* is

$$H_P = \hbar\omega_0 \log \hat{N} = \hbar\omega_0 \sum_{p \in \text{Pr}} (\log p) \hat{n}_p, \quad \omega_0 > 0, \quad (6)$$

with domain

$$D(H_P) = \left\{c \in \ell^2(\mathbb{N}_{\geq 1}) : \sum_{n \geq 1} (\hbar\omega_0 \log n)^2 |c_n|^2 < \infty\right\}. \quad (7)$$

The two expressions in (6) agree on the finite-support domain, on which they are essentially self-adjoint; the closure is the self-adjoint operator with domain (7).

Equation (6) exhibits H_P as a *free Bose gas* whose one-particle spectrum is $\{\log p : p \in \text{Pr}\}$: each prime is one species of elementary excitation (a *primon*), and a many-body state $|n\rangle$ has energy equal to the logarithm of the integer it represents. This is Julia's primon gas [1], cleaned of its informal packaging.

Theorem 2.4 (Self-adjointness, spectrum, and simplicity). H_P is self-adjoint on $D(H_P)$ and nonnegative, with compact resolvent. Its spectrum is the pure point set

$$\text{spec}(H_P) = \overline{\{\hbar\omega_0 \log n : n \in \mathbb{N}_{\geq 1}\}}, \quad (8)$$

a discrete set whose only accumulation point is $+\infty$; it has no finite accumulation point and no continuous spectrum. Every eigenvalue $\hbar\omega_0 \log n$ is simple: $\log m = \log n$ if and only if $m = n$. The gaps between consecutive levels are $\hbar\omega_0 \log(1 + \frac{1}{n})$, which tend to 0: the levels densify toward $+\infty$ with unit degeneracy throughout. The ground state is the vacuum $|1\rangle$ with eigenvalue 0, and it is unique.

Proof. Under the unitary $U : \mathcal{H}_{\mathcal{A}} \rightarrow \ell^2(\mathbb{N}_{\geq 1})$, $U|k\rangle = |n\rangle$ (unitary by the fundamental theorem of arithmetic), H_P becomes multiplication by the real sequence $a_n = \hbar\omega_0 \log n$. Multiplication by a real-valued sequence is self-adjoint on its maximal domain $\{c \in \ell^2 : \sum_n a_n^2 |c_n|^2 < \infty\}$, and its spectrum is the closure of $\{a_n\}$. Since $a_n \rightarrow \infty$ monotonically, the closure adds no finite points. Simplicity: if $a_m = a_n$ with $m \neq n$, then $\log m = \log n$ forces $m = n$, a contradiction. The ground state: $a_n \geq 0$ with equality only at $n = 1$. $\square$

Two consequences deserve emphasis. First, the *unique zero-entropy ground state* $|1\rangle$ is the multiplicative identity; this is the arithmetic floor underlying the Janus point developed in Sections 6 and 9. Second, the *simplicity* of every level means the primon spectrum has no degeneracy whatsoever: the integers label states one-to-one. Any degeneracy in a putative physical realization must therefore come from sectors *beyond* H_P (for instance the Gaussian sector of Section 8).

2.3 The partition function and the Hagedorn temperature

Proposition 2.5 (Partition function). *For inverse temperature $\beta > 0$,*

$$Z(\beta) = \text{Tr } e^{-\beta H_P} = \sum_{n=1}^{\infty} e^{-\beta \hbar \omega_0 \log n} = \sum_{n=1}^{\infty} n^{-\beta \hbar \omega_0} = \zeta(\beta \hbar \omega_0), \quad (9)$$

and the series converges if and only if $\beta \hbar \omega_0 > 1$, i.e. $T < T_H$, where

$$T_H = \frac{\hbar \omega_0}{k_B} \quad (10)$$

is the Hagedorn temperature of the primon gas. On the half-line $\sigma = \beta \hbar \omega_0 > 1$, the Euler product

$$\zeta(\sigma) = \prod_{p \in \text{Pr}} \frac{1}{1 - p^{-\sigma}} = \prod_{p \in \text{Pr}} \frac{1}{1 - e^{-\beta \varepsilon_p}}, \quad \varepsilon_p := \hbar \omega_0 \log p, \quad (11)$$

is precisely the grand-canonical factorization of a free Bose gas into independent prime modes, with each mode contributing the standard Bose factor. The mean occupation of mode p is

$$\langle \hat{n}_p \rangle_{\beta} = \frac{1}{e^{\beta \varepsilon_p} - 1} = \frac{1}{p^{\beta \hbar \omega_0} - 1}. \quad (12)$$

Proof. The trace formula is the spectral decomposition of Theorem 2.4; convergence at $\sigma > 1$ and the Euler product are Euler's theorems; the occupation formula is the standard free-boson computation on each mode. $\square$

The identification $Z = \zeta$ carries a consequence that gives it physical teeth: the primon gas is a *Hagedorn system* [10]. Its counting function grows exponentially in the energy, and the partition function diverges at a finite critical temperature. In string theory this is the famous signal that, near T_H , energy input goes into multiplying the number of light degrees of freedom rather than into heating. The same reading applies here exactly, with one structural difference worth recording: every primon level has *unit degeneracy* (Theorem 2.4), so the exponential density of states $\rho(E) = \sum_n \delta(E - \hbar \omega_0 \log n)$, of size $e^{E/\hbar \omega_0}/\hbar \omega_0$, is produced entirely by level *densification* (gap $\hbar \omega_0 \log(1 + 1/n) \rightarrow 0$), not by degeneracy growth. The primon gas is the degeneracy-one limiting case of a Hagedorn system: the exponential growth rate is exactly $1/\hbar \omega_0$ and nothing multiplies it. Since the Gibbs state exists only for $\beta \hbar \omega_0 > 1$, all statements at or above T_H are microcanonical or cutoff statements, never canonical expectation values; the clock family of Section 5 is the explicit analytic object that lives on both sides of that line.

Theorem 2.6 (Hagedorn asymptotics). *Let $k_B = 1$ and write $T_H = \hbar \omega_0$. As $T \uparrow T_H$ (i.e. $\sigma = \beta \hbar \omega_0 \downarrow 1$):*

$$\zeta(\sigma) = \frac{1}{\sigma - 1} + \gamma + O(\sigma - 1), \quad (13)$$

$$U(T) := -\frac{\partial}{\partial \beta} \log Z(\beta) = \frac{\hbar \omega_0}{\sigma - 1} + O(\hbar \omega_0) = \frac{T T_H}{T_H - T} + O(T_H), \quad (14)$$

$$S(T) := \beta U + \log Z = \frac{1}{\sigma - 1} - \log(\sigma - 1) + O(1), \quad (15)$$

$$C(T) := \frac{dU}{dT} = \frac{T_H^2}{(T_H - T)^2} (1 + O((T_H - T)/T)). \quad (16)$$

In particular the specific heat diverges as $T \uparrow T_H$, while the entropy diverges only as $T/(T_H - T)$; energy is absorbed at nearly constant temperature.

Proof. The Laurent expansion $\zeta(\sigma) = 1/(\sigma - 1) + \gamma + O(\sigma - 1)$ is classical [11]. Then $-\zeta'/\zeta(\sigma) = [1/(\sigma - 1)^2 + O(1)]/[1/(\sigma - 1) + O(1)] = 1/(\sigma - 1) + O(1)$, giving $U = \hbar\omega_0(-\zeta'/\zeta)(\sigma)$. Substituting $\sigma = T_H/T$ gives $\sigma - 1 = (T_H - T)/T$ and the stated form. The entropy follows from $S = \beta U + \log Z$ with $\log Z = -\log(\sigma - 1) + O(1)$. Differentiating $U = TT_H/(T_H - T)$ in T gives $C = T_H^2/(T_H - T)^2$ up to the stated relative error. $\square$

Proposition 2.7 (Ensemble bridge: the pole is the counting law). *For $\sigma > 1$ the canonical partition function is exactly the Laplace–Stieltjes transform of the microcanonical counting function N of Theorem 6.1 (with $x = E/\hbar\omega_0$):*

$$Z(\sigma) = \zeta(\sigma) = \int_0^\infty e^{-\sigma x} dN(x) = \sum_{n \geq 1} n^{-\sigma}. \quad (17)$$

Moreover:

1. the pole part of Z is exactly the Laplace transform of the exponential counting law: $\int_0^\infty e^{-(\sigma-1)x} dx = 1/(\sigma - 1)$;
2. $Z(\sigma) - 1/(\sigma - 1)$ extends to an entire function [11], and tends to the Euler constant γ as $\sigma \downarrow 1$.

The canonical divergence at the Hagedorn point is therefore not an ensemble artifact set against the microcanonical counting: the two are related by an exact integral transform, and the pole residue 1 is the microcanonical growth rate. The Hagedorn temperature is a property of the counting law itself, read on either transform.

Proof. The Stieltjes measure of the staircase $x \mapsto \lfloor e^x \rfloor$ is $\sum_{n \geq 1} \delta_{\log n}$, so the integral in (17) equals $\sum_n e^{-\sigma \log n} = \zeta(\sigma)$ for $\sigma > 1$ (absolute convergence). (i) is elementary calculus. (ii) is the classical removal of the unique singularity of ζ at $s = 1$ [11]; the constant term γ of the Laurent expansion appears in Theorem 2.6. $\square$

Remark 2.8 (Microcanonical reading). Section 6 proves the counting law $N(E) = \lfloor e^{E/\hbar\omega_0} \rfloor$ with a defect that decays like $e^{-E/\hbar\omega_0}$: the exponential growth rate of the level count is exactly $1/\hbar\omega_0$, uniformly in E . Equivalently, the level density grows as $e^{E/\hbar\omega_0}/\hbar\omega_0$, so every local average of $\log \rho$ over a fixed energy window has slope $1/\hbar\omega_0 + O(e^{-E/\hbar\omega_0})$: energy input converts into state multiplicity at a fixed exchange rate, at every energy scale. (The unsmoothed entropy $k_B \log N(E)$ is a staircase: its pointwise derivative vanishes between levels, and the statement just made concerns the growth rate of fixed-window smoothings, not the pointwise derivative.) This is the sharpened form of the claim that “the state space expands exponentially”: the expansion rate is not merely large, it is *uniform* in E , and that uniformity is what the pole of ζ at $s = 1$ encodes—and Proposition 2.7 makes that encoding an exact transform identity. The classical smooth term “ x ” of the prime number theorem is the same statement read on the counting function $\psi(x)$, and the oscillatory remainder $-\sum_\rho x^\rho/\rho$ is the fluctuation sector; the paper returns to this duality in Sections 5 and 8.

2.4 Gaussian extension

The arithmetic sector extends to the Gaussian integers $\mathbb{Z}[i]$ for the putative five-dimensional, chirality-carrying sector. The mathematics of the extension itself is exact and standard—it is the theory of the number field $\mathbb{Q}(i)$ —and it supplies the clean statement that any dimensional or chirality reading must lean on: what the extension does *not* supply is the dimension count or a chirality operator, both of which are dictionary entries (C9, C10) and are labeled as such.

Definition 2.9 (Gaussian primon gas). Let $\mathcal{P}_{\mathbb{Z}[i]}$ denote the Gaussian primes modulo units $\{\pm 1, \pm i\}$, with norm $N(\pi) = |\pi|^2$. The *Gaussian arithmetic Hilbert space* is $\mathcal{H}_{\mathbb{Z}[i]} = \ell^2(\mathcal{F}_{\mathbb{Z}[i]})$ over the free commutative monoid on $\mathcal{P}_{\mathbb{Z}[i]}$, and the Gaussian primon Hamiltonian is

$$H_{\mathbb{Z}[i]} = \hbar\omega_0 \sum_{\pi \in \mathcal{P}_{\mathbb{Z}[i]}} \log N(\pi) \hat{n}_\pi, \quad Z_{\mathbb{Z}[i]}(\beta) = \prod_{\pi} \frac{1}{1 - N(\pi)^{-\beta\hbar\omega_0}}. \quad (18)$$

Theorem 2.10 (Dedekind factorization). $Z_{\mathbb{Z}[i]}(\beta) = \zeta_{\mathbb{Q}(i)}(\beta\hbar\omega_0)$, the Dedekind zeta function of the Gaussian rationals, and

$$\zeta_{\mathbb{Q}(i)}(s) = \zeta(s) L(s, \chi_{-4}), \quad (19)$$

where $L(s, \chi_{-4}) = \sum_m (-1)^{m-1} (2m-1)^{-s}$ is the Dirichlet L -function of the nontrivial character mod 4. Both factors converge for $\sigma > 1$; the product is entire on the common half-plane and $L(1, \chi_{-4}) = \pi/4 \neq 0$.

Proof. Unique factorization of ideals in $\mathbb{Z}[i]$ (it has class number 1) gives the Euler product over Gaussian prime ideals; identifying prime ideals with Gaussian primes modulo units gives the stated product, and the splitting classification of rational primes in $\mathbb{Z}[i]$ ($p \equiv 1 \pmod{4}$ splits, $p \equiv 3 \pmod{4}$ is inert, 2 ramifies) yields $\zeta_{\mathbb{Q}(i)} = \zeta L(\chi_{-4})$ [12]. The nonvanishing $L(1, \chi_{-4}) = \pi/4$ is Leibniz's series. $\square$

Equation (19) exhibits the Gaussian sector as the rational sector *times* an asymmetric factor: $L(s, \chi_{-4})$ is precisely the asymmetry between the splitting class $p \equiv 1 \pmod{4}$ and the inert class $p \equiv 3 \pmod{4}$ —a binary, with exact statistics. In the dictionary this binary is read as chirality; the reading is conjectural, not structural: the units $\{\pm 1, \pm i\}$ are quotiented out in the norm $N(\pi)$, so no orientation survives at the level of norms, and no spacetime dimension is selected by the factorization. This is developed into a precise (and appropriately demoted) arithmetic model of charge conjugation and CP violation in Section 8.

2.5 What the arithmetic sector does and does not assert

It is worth fixing the epistemic boundary before dynamics is introduced. Everything in this section is unconditional mathematics: the primon gas exists as a quantum system, its spectrum is exactly the logarithms of the integers, its thermodynamics is exactly the analytic theory of ζ , and its critical behavior is exactly Hagedorn behavior. *No claim has yet been made that any physical system is a primon gas.* The identification $Z_{\text{grav}}^{\text{ren}}(s) = \zeta(s)$ for near-singularity gravity is Conjecture C1 of the register, the reported 2025–2026 developments [3] being its current physical evidence. The discipline of this paper is that such identifications live in the register, not in the theorems.

The boundary is worth drawing on the other side as well, because each of the following inferences is tempting and none is free: unique factorization does not imply holography; the identity $Z = \zeta$ does not imply a black hole; $\mathbb{Q}$ -linear independence of the prime logarithms does not imply a beginning of time; the Dedekind factorization (19) does not imply a fifth dimension or a chiral fermion; and level statistics on truncated occupation lattices do not, by themselves, imply that arithmetic thermalizes. Each of these steps requires a register entry with its own evidence and falsifier, and the theorems of this section license none of them.

3 Dynamics: The Interaction Sector on the Prime Lattice

3.1 The frozen Hamiltonian

The dynamical claims of the framework—transitions into composite integer states, and factorization of the evolving state into highly composite integers—require a Hamiltonian with off-diagonal matrix elements in the occupation basis. The primon Hamiltonian has none:

Proposition 3.1 (Frozen spectrum). $[H_P, \hat{N}] = 0$ and $[H_P, \hat{n}_p] = 0$ for every p . Consequently every basis vector $|n\rangle$ is a stationary state: $e^{-i\tau H_P/\hbar}|n\rangle = e^{-i\omega_0\tau \log n}|n\rangle$ never leaves the ray $\mathbb{C}|n\rangle$. In particular the vacuum $|1\rangle$ is an eigenstate with eigenvalue 0 and evolves trivially, $e^{-i\tau H_P/\hbar}|1\rangle = |1\rangle$ for all τ .

Proof. Both operators are diagonal in the same basis (Theorem 2.4). $\square$

A diagonal Hamiltonian cannot create primons, destroy them, or entangle anything. The “arithmetic cascade,” the entanglement growth, and the emergence of a thermodynamic arrow all require dynamics *off* the diagonal of $\hat{N}$. The interaction sector must be supplied. Meeting this requirement produces, as a byproduct, the objects on which the rest of the framework rests: the multiplicative shift operators, the prime lattice, and a well-posed thermalization problem.

3.2 Multiplicative shift operators

Definition 3.2 (Multiplication operator). For $n \in \mathbb{N}_{\geq 1}$ define $M_n: \mathcal{H}_{\mathcal{A}} \rightarrow \mathcal{H}_{\mathcal{A}}$ on basis vectors by

$$M_n|k\rangle = |nk\rangle, \quad (20)$$

extended linearly, with adjoint $M_n^\dagger|k\rangle = |k/n\rangle$ if $n \mid k$ (i.e. $k_p \geq a_p$ for all p with $n = \prod p^{a_p}$) and $M_n^\dagger|k\rangle = 0$ otherwise.

Proposition 3.3 (Monoid representation). The assignment $n \mapsto M_n$ is an isometric representation of the multiplicative monoid $(\mathbb{N}_{\geq 1}, \times)$ on $\mathcal{H}_{\mathcal{A}}$:

1. $M_m M_n = M_{mn} = M_n M_m$ (the representation is abelian);
2. $M_n^\dagger M_n = I$ (each M_n is an isometry);
3. $M_n M_n^\dagger = I - P(n \mid k)$, where $P(n \mid k)$ is the projection onto $\overline{\text{span}}\{|k\rangle : n \mid k\}$, a product over the modes $p^{a_p} \parallel n$ of the projections onto $\{\hat{n}_p \geq a_p\}$;
4. $\|M_n\| = \|M_n^\dagger\| = 1$, and $M_1 = I$.

Moreover, for each prime p ,

$$M_p = a_p^\dagger (\hat{n}_p + 1)^{-1/2}, \quad (21)$$

so M_p is the unilateral shift of the p -mode tensored with the identity on all other modes.

Proof. (1) $M_m M_n|k\rangle = |mnk\rangle$ by associativity and commutativity of multiplication in $\mathbb{N}$. (2) M_n maps the orthonormal basis $\{|k\rangle\}$ to the orthonormal family $\{|nk\rangle\}$ (distinct k give distinct nk by cancellation in $\mathbb{N}$), hence is an isometry. (3) The initial space of $M_n^\dagger$ is the range of M_n , i.e. the closed span of $|k\rangle$ with $n \mid k$; this condition factorizes across modes because divisibility does. (4) is immediate. For (21): on $|k\rangle$, $a_p^\dagger (\hat{n}_p + 1)^{-1/2}|k\rangle = \sqrt{k_p + 1} (k_p + 1)^{-1/2}|k + e_p\rangle = |k + e_p\rangle = M_p|k\rangle$. $\square$

The operators M_n are the arithmetic analogues of the Weyl shifts, and Proposition 3.3 is the exact statement behind the phrase that composites are “correlated multi-prime states”: multiplication *creates* occupation, division *annihilates* it, and the two are adjoints. Note also the co-isometric defect $M_n M_n^\dagger = I - P(n \mid k)$: multiplication by n is invertible precisely on states already divisible by n . The failure of unitarity of each individual M_n is the operator-theoretic fingerprint of the irreversibility that arithmetic carries: the semigroup $(\mathbb{N}, \times)$ is free, hence has cancellation but no inverses, and the quantum representation inherits exactly that structure. The commutator with the free Hamiltonian is exact and arithmetic: on the finite-support domain,

$$[H_P, M_m] = \hbar\omega_0(\log m) M_m, \quad (22)$$

so each multiplicative shift is an eigenoperator of the derivation $[H_P, \cdot]$ with eigenvalue exactly $\hbar\omega_0 \log m$: multiplication by m costs $\hbar\omega_0 \log m$ in energy, and nothing else. This identity is the resonance structure that the interaction sector must work with (and the origin of the Wannier–Stark suppression analyzed in Remark 3.12).

3.3 The walk Hamiltonian and the prime lattice

Define, for a finite set $P_f \subset \text{Pr}$ of active modes and real couplings $g = (g_p)_{p \in P_f}$, the *multiplicative walk Hamiltonian*

$$H_W(P_f, g) = \sum_{p \in P_f} g_p (M_p + M_p^\dagger), \quad (23)$$

and the *hopping Hamiltonian*

$$H_{\text{hop}}(P_f, h) = \sum_{p \neq q \in P_f} h_{pq} (a_p^\dagger a_q + a_q^\dagger a_p). \quad (24)$$

Proposition 3.4 (Well-posed interacting dynamics). *For finite P_f , H_W is bounded and self-adjoint, and the number-conserving part $H_P + H_{\text{hop}}$ is essentially self-adjoint on the finitely supported configurations (Proposition 3.9). The full Hamiltonian*

$$H_{\text{tot}} = H_P + H_W + H_{\text{hop}} \quad (25)$$

is self-adjoint on the domain of the closure of $H_P + H_{\text{hop}}$ (Kato–Rellich, the bounded H_W being a relative-bound-zero perturbation), and $[H_{\text{tot}}, \hat{N}] \neq 0$: the dynamics generates genuine transitions

$$\langle m | H_W | k \rangle \neq 0 \iff m/k \in \{p, p^{-1}\} \text{ for some } p \in P_f, \quad (26)$$

i.e. precisely the edges of the divisibility graph of $\mathbb{N}$ restricted to P_f .

Proof. Each M_p is bounded with norm 1 (Proposition 3.3); a finite sum of bounded self-adjoint operators is bounded self-adjoint, so H_W is bounded. The hop $a_p^\dagger a_q$ is number-conserving but *unbounded* on $\ell^2(\mathcal{F})$ (its matrix elements $\sqrt{(k_p + 1)k_q}$ grow linearly with the occupation), so it is not a bounded perturbation of H_P ; its self-adjointness is instead a consequence of the occupation-sector decomposition of Proposition 3.9, which applies verbatim to $H_P + H_{\text{hop}}$. H_{tot} is then a bounded perturbation of the self-adjoint closure of $H_P + H_{\text{hop}}$, hence self-adjoint on its domain. Non-commutation: M_p changes the eigenvalue of $\hat{N}$ from n to np , and a direct computation gives

$$[H_W, \hat{N}]|n\rangle = \sum_p g_p n \left(-(p-1)|np\rangle + \left(1 - \frac{1}{p}\right) \mathbf{1}_{p|n} |n/p\rangle \right) \neq 0 \quad (27)$$

for $n \geq 1$ and nonzero g : the two coefficients are different in both sign and magnitude ($-(p-1)$ versus $1 - 1/p$), so the terms cannot cancel. $\square$

Three remarks interpret these objects.

Remark 3.5 (The prime lattice). H_{hop} is a Bose–Hubbard-type tight-binding Hamiltonian on a lattice whose *sites* are the primes, with on-site energies $\varepsilon_p = \hbar\omega_0 \log p$ and hopping h_{pq} . We call this graph the *prime lattice*. It is the natural home for every interaction claim in the framework: interactions among prime modes are hops along the lattice; the chaotic sector conjectured in Conjecture C5 is a spectral-statistics statement about H_{tot} on the prime lattice; and the “chemistry of the primes” becomes the statement that the Standard Model’s force sectors correspond to structured sub-lattices (Conjecture C9). A boundary note on provenance: the Hamiltonians (23)–(24) and the interacting completions of Conjecture 3.1 are *selected*, not derived. What the M_n algebra supplies is the exact energy bookkeeping (22) and the graph

(the divisibility edges of H_W , the exponent-lattice edges of H_{hop}); the choice of couplings and of the quartic completions is fixed by symmetry requirements—boundedness, self-adjointness, $\hat{N}_{\text{tot}}$ -preservation, and the breaking of the free integrability—and each candidate is then a conjecture’s object whose thermalization is an empirical question, decided numerically in Remark 3.12.

Remark 3.6 (Conservation structure). The three sectors carry a strict conservation hierarchy. H_P is diagonal in the occupation basis and conserves everything. H_{hop} moves single primons between modes ($a_p^\dagger a_q$: one occupation up, one down), so it conserves the *total primon number* $\hat{N}_{\text{tot}} = \sum_p \hat{n}_p$ exactly, but not $\hat{N}$ (it rescales n by q/p). H_W conserves *neither*: each M_p raises the occupation of the p -mode by one and hence raises $\hat{N}_{\text{tot}}$ by one, while each $M_p^\dagger$ lowers it—the walk is the nucleation sector, the engine of entropy growth in the precise quantitative sense of Proposition 3.13 ($d\langle \hat{N}_{\text{tot}} \rangle / d\tau > 0$ from $|1\rangle$ whenever $H_W \neq 0$). The full theory therefore has a graded conservation hierarchy: H_P conserves $\hat{N}$ and $\hat{N}_{\text{tot}}$; H_{hop} conserves $\hat{N}_{\text{tot}}$ only; H_W conserves neither and is the only source of net primon creation. The additive and multiplicative structures of $\mathbb{N}$ thereby acquire distinct dynamical roles—spectral order (H_P), transport (H_{hop}), creation (H_W)—which separates “spectral order” from “arithmetic cascade.” Register note: the CP ratchet of C10 lives entirely in this non-conserving H_W sector.

Remark 3.7 (Relation to the reported physics). The conformal primon gas reported near black-hole singularities [3] corresponds in this language to the claim that *some* sector of near-singularity gravity is governed by H_P (or its Gaussian analogue) with the conformal $\mathfrak{sl}(2, \mathbb{R})$ symmetry of the axioms acting on the ω_0 scale. The interacting extension (25) is the framework’s addition; its necessity follows from the arrow-of-time claims, which (by Proposition 3.1) are impossible under H_P alone.

3.4 The cascade: statics and dynamics

The “thermodynamic cascade” is the framework’s central image from sparse prime states into dense composite states. As dynamics under H_P this is impossible (Proposition 3.1); as a statement about the *microcanonical ensemble*, however, it is a classical theorem. The dichotomy therefore splits the cascade into a theorem (statics) and a conjecture (dynamics).

Theorem 3.8 (Normal order and Erdős–Kac for the primon ensemble). *Fix $E > 0$ and let $x = e^{E/\hbar\omega_0}$. Equip $\{n \leq x\}$ with the uniform measure—the cumulative microcanonical ensemble of H_P at energy E (the shell ensemble on $\{x(1-\delta) < n \leq x\}$ has the same normal order for every fixed $\delta \in (0, 1)$, since the normal-order bound holds on every dyadic range; the Gaussian limit is stated for the cumulative measure). Let $\omega(n)$ be the number of distinct prime factors of n . Then:*

1. (Hardy–Ramanujan [13]; Turán [14].) $\omega(n)$ has normal order $\log \log n$: for every $\varepsilon > 0$ the proportion of $n \leq x$ with $|\omega(n) - \log \log x| > \varepsilon \log \log x$ tends to 0 as $x \rightarrow \infty$.
2. (Erdős–Kac [15].) The distribution of $(\omega(n) - \log \log x) / \sqrt{\log \log x}$ converges to the standard Gaussian: for all t ,

$$\lim_{x \rightarrow \infty} \frac{1}{x} \# \left\{ n \leq x : \frac{\omega(n) - \log \log x}{\sqrt{\log \log x}} \leq t \right\} = \Phi(t). \quad (28)$$

The interpretation is exact. The number of states at energy E is $\lfloor e^{E/\hbar\omega_0} \rfloor$ (Section 6), and Theorem 3.8 describes the *typical* member of that ensemble: a composite integer with about $\log \log x$ distinct prime factors, $\log \log x = \log(E/\hbar\omega_0)$ growing only logarithmically in the energy. The entropy $\log N(E) = E/\hbar\omega_0 + O(1)$ is astronomically larger than the typical complexity $\log \log N(E)$: the ensemble is dominated by integers that are composite but *far from maximally* composite (the maximally composite integers, the powers of two and their smooth kin,

are exponentially atypical). Thus a “cascade into composite states” is correct as a description of where the measure lives, and the Erdős–Kac Gaussian is the precise fluctuation law of the cascade. What remains open is the dynamical version:

Conjecture 3.1 (Cascade/thermalization on the prime lattice, interacting completion). *Let*

$$H_{\text{tot}}^{(U,V)} = H_P + H_W + H_{\text{hop}} + U \sum_{p < q} \hat{n}_p \hat{n}_q + V \sum_{p < q} (\hat{n}_p + \hat{n}_q) (a_p^\dagger a_q + a_q^\dagger a_p) \quad (29)$$

on $\mathcal{H}_A$ with generic couplings and $U \neq 0$ or $V \neq 0$ (diagonal or kinetic interacting completion; both are quartic in the ladder operators, and the kinetic one is off-diagonal and $\hat{N}_{\text{tot}}$ -preserving, with matrix elements $V(k_p + k_q)\sqrt{(k_p + 1)k_q}$ on the hops of the prime lattice), restricted to the energy window $[E, E + \Delta]$. Then along any truncation sequence with truncation dimension $D \rightarrow \infty$ at fixed effective dose: (i) eigenstate expectation values of local occupation observables are smooth in E with within-window fluctuations $\sigma_{\text{ETH}}(D) \rightarrow 0$; (ii) diagonal-ensemble values converge to the microcanonical values of Theorem 3.8; and (iii) the relaxation time scale is polynomial in E . The bare H_{tot} ($U = V = 0$) is excluded: it is a quasi-quadratic object (see below) to which the thermalization question does not apply.

Conjecture 3.1 (register entry C4) is the engine for every arrow-of-time claim in the framework: entropy growth, decoherence, record formation, and the emergence of the second law are all downstream of it. It is a well-posed, hard, and conventional-sounding problem of quantum statistical mechanics, which is exactly the point: the framework’s most sweeping physical claims reduce to questions that experts know how to test. The two completions in (29) are not equivalent and the difference matters: the U term is a diagonal potential on the exponent lattice (a Wannier–Stark-type renderer, as the data confirm), while the V term is the Bose–Hubbard-type *density-assisted hopping* known from the interacting-boson literature—quartic yet kinetic. Before the numerical program, the well-posedness of the kinetic completion itself:

Proposition 3.9 (Occupation sectors; self-adjointness of the completions). *Fix a finite active set P_f (d modes), real couplings h , U , V , and let $\mathcal{H}_A = \ell^2(\mathcal{F})$.*

1. The number-conserving part $H_{\text{nc}}^{(U,V)} = H_P + U \sum_{p < q} \hat{n}_p \hat{n}_q + H_{\text{hop}} + V \sum_{p < q} (\hat{n}_p + \hat{n}_q) (a_p^\dagger a_q + a_q^\dagger a_p)$ commutes with $\hat{N}_{\text{tot}}$ and with $\hat{n}_q$ for every $q \notin P_f$: it decomposes as an orthogonal direct sum of finite-dimensional blocks $H^{(\kappa, S)}$, indexed by the occupations κ outside P_f and the total occupation S inside P_f , of dimension $\binom{S+d-1}{d-1}$. Each block is Hermitian; for the kinetic term this is the bond-reversal invariance of its weight: a hop raises k_p by one and lowers k_q by one, so $(\hat{n}_p + \hat{n}_q)$ commutes with $a_p^\dagger a_q$ and the matrix elements $V(k_p + k_q)\sqrt{(k_p + 1)k_q}$ are symmetric across every bond. The sector norms obey the row-sum bound $\|H_{\text{kin}}^{(S)}\| \leq |V|(d-1)S(S+1)$.
2. $H_{\text{nc}}^{(U,V)}$ is essentially self-adjoint on the finitely supported configurations; its closure is the direct sum of the blocks with maximal domain $\{\psi : \sum_{\kappa, S} \|H^{(\kappa, S)}\psi_{\kappa, S}\|^2 < \infty\}$. Since H_W is bounded (Proposition 3.4), the full interacting completion $H_{\text{tot}}^{(U,V)}$ of (29) is self-adjoint on that domain for every $U, V \in \mathbb{R}$ (Kato–Rellich, relative bound zero): no smallness condition on the couplings is needed.
3. Box truncations. Let $H^{(K)}$ be the restriction of $H_{\text{tot}}^{(U,V)}$ to the box $\{k_p \leq K\}$. Every sector with $S \leq K$ lies entirely inside the box, and the sectors are invariant under the number-conserving part, so for $H_W = 0$ the truncated dynamics is exact on each such sector,

$$e^{-i\tau H^{(K)}} \psi = e^{-i\tau H_{\text{nc}}^{(U,V)}} \psi,$$

for every state ψ supported in a sector $S \leq K$ and all τ . With $H_W \neq 0$ the sectors mix through the walk, and the truncated propagators converge to the untruncated one strongly, uniformly on compact τ -intervals: the finitely supported configurations are a core for $H_{\text{tot}}^{(U,V)}$ on which the truncations stabilize, and Trotter–Kato transfers the convergence to the resolvents and hence to the propagators.

Proof sketch. (i) The conservation identities are immediate from the ladder algebra; finiteness of the sectors is the finiteness of $\{k \in \mathbb{N}^d : \sum k_p = S\}$; Hermiticity and the norm bound follow from the displayed matrix elements (each row of a sector block carries at most $2\binom{d}{2}$ bonds of weight at most $(k_p + k_q)(k_p + k_q + 1)/2 \leq S(S+1)/2$, and $\sum_{p < q} (k_p + k_q) \leq (d-1)S$). (ii) A direct sum of finite Hermitian blocks is self-adjoint on its maximal domain, and the algebraic direct sum of the block spaces—the finitely supported configurations—is a core. (iii) For $H_W = 0$ the block containing ψ is contained in the box, both propagators act as the same finite matrix on it, and the orthogonal blocks are untouched. For the general case, each term maps the finitely supported configurations into themselves, so $H^{(K)}\psi = H\psi$ once K exceeds the support of ψ ; Trotter–Kato (the sequence of self-adjoint truncations with uniformly bounded resolvents converging on a core) yields strong resolvent convergence and hence the uniform strong convergence of $e^{-i\tau H^{(K)}}$ on compact τ -intervals. $\square$

The sector structure is corroborated numerically at the assembly level: every extracted sector block is Hermitian to machine precision and satisfies the norm bound for $S \leq 12$ at $d = 3, 4$; the box-exactness of sector-supported dynamics holds to 7×10^{-15} across truncations, while the walk control ($g \neq 0$) leaks out of the sector exactly as H_W ’s non-conservation requires (Remark 3.6).

Proposition 3.10 (Sector Jacobi structure). *Fix $d \geq 2$, total occupation S , $\kappa = 0$, and let $H^{(S)}$ be the block of $H_{\text{nc}}^{(U,V)}$ on the compositions of S into d parts.*

1. Layer grading. *Order the compositions by the layer $\ell = S - k_1$, the hop distance from the corner $(S, 0, \dots, 0)$. Every number-conserving hop either transfers one quantum between modes $2, \dots, d$ (ℓ unchanged) or exchanges one quantum with mode 1 ($\ell \rightarrow \ell \pm 1$); in this grading $H^{(S)}$ is exactly block-tridiagonal, and the diagonal block at layer ℓ is the $(d-1, \ell)$ sector of the mode family $\{p_2, \dots, p_d\}$ (same h, V), shifted by the affine term $(S - \ell)(\log p_1 + U\ell)$. The recursion terminates at $d = 2$, where the sector is the Jacobi matrix*

$$a_k = k \log p_1 + (S - k) \log p_2 + U k(S - k), \quad b_k = (2h_{12} + VS) \sqrt{(k+1)(S-k)},$$

$$k = 0, \dots, S.$$

2. Two-mode solvability. *At $U = 0$ the two-mode block is the rotated spin problem $\Delta \hat{S}_z + 2C \hat{S}_x + \frac{S}{2} \log(p_1 p_2)$ with $\Delta = \log(p_1/p_2)$, $C = 2h_{12} + VS$, in the spin- $S/2$ representation. Its spectrum is the picket fence $E_j = \frac{S}{2} \log(p_1 p_2) + \sqrt{\Delta^2 + 4C^2} (\frac{S}{2} - j)$, its eigenvectors are Krawtchouk polynomials in k , and the eigenstate occupation is exactly linear in the eigenvalue, $\langle k_1 \rangle_j = \frac{S}{2} + (\frac{S}{2} - j)\Delta/\sqrt{\Delta^2 + 4C^2}$. Under the full-spectrum 30-bin protocol the fluctuation ratio therefore converges to the S -independent floor $|\Delta|/(30\sqrt{\Delta^2 + 4C^2})$, while the Haar benchmark decays as $n^{-1/2}$: the fluctuation floor does not decay. At fixed V the floor itself decays as $1/S$; the picket ratio statistic $\langle r \rangle = 1$ and the exact linearity survive at every scale.*
3. Free reduction and permutation covariance. *H_P and H_{hop} are one-body operators: their sector blocks are the S -fold symmetric representations of a fixed $d \times d$ Hermitian matrix, diagonalizable once and for all at every S . The kinetic term is the only interaction in the sector problem. At matched dose $VS^2 \approx 36$ its share of the sector bandwidth decays as $1/S$ (the sector ladder approaches the free limit); at fixed V it dominates the bandwidth, consistent with the row-sum bound of Proposition 3.9. The kinetic interaction is exactly covariant under the full mode-permutation group: at $h = 0$ its commutator with every mode transposition*

and cycle vanishes identically, and the prime-logarithm diagonal is the term that breaks the covariance, at relative strength $O(1/(VS))$ at fixed V .

Proof sketch. (i) A hop $a_p^\dagger a_q$ moves one quantum from q to p ; if neither index is 1 the occupation k_1 and hence ℓ are unchanged, and if one index is 1 the layer changes by one, so the grading is block-tridiagonal; the diagonal blocks retain the $(d-1, \ell)$ sector of the remaining modes because the internal couplings and the kinetic bond weights $(k_p + k_q)\sqrt{(k_p + 1)k_q}$ do not involve k_1 beyond the affine diagonal contribution $(S - \ell)\log p_1 + U(S - \ell)\ell$. At $d = 2$ the layers are singletons and the block is tridiagonal with the stated entries (the symmetrized hop contributes $2h_{12}$ per bond, the kinetic bond weight $(k_1 + k_2)\sqrt{(k_1 + 1)k_2} = S\sqrt{(k + 1)(S - k)}$). (ii) With $U = 0$ the diagonal is affine in the spin magnetization and the off-diagonal is the S_+ matrix element, so a rotation of the spin quantization axis diagonalizes the block; the stated spectrum, eigenvectors, and occupation linearity follow, and the bin-width fluctuation of a linear observable–energy relation is $|d\langle k_1 \rangle/dE|$ times the bin width over $\sqrt{12}$. (iii) The one-body algebra and the symmetric representation are standard; the covariance follows because the kinetic sum runs over unordered mode pairs with pair-independent coupling, and permutations relabel pairs; the diagonal distinguishes the primes. $\square$

The proposition is validated numerically to roundoff: the assembly of $H^{(S)}$ from the compositions equals the full-grid extraction at $K = S$ (deviations $\leq 4 \times 10^{-15}$ at $d = 2, 3, 4$ in the U, V cross-checks), the block-tridiagonal pattern holds with no coupling beyond $|\Delta\ell| = 1$ at every tested (d, S) , the diagonal-block recursion matches the shifted sub-sector to 3.6×10^{-15} , and the two-mode closed forms reproduce the tridiagonal eigensolve to $\leq 4 \times 10^{-13}$ at $S \leq 512$ in both dose conventions. The sector ladders built on this structure, and the classical limits they select, are the mathematical falsifier program for C4 (Remark 3.12(1); Section 11.3, item 1).

Remark 3.11 (The mode-count limit is singular). For an infinite active set the interacting terms do not define operators on $\ell^2(\mathcal{F})$ at all: from any configuration with $k_q \geq 1$, the hop $a_p^\dagger a_q$ reaches every unoccupied mode $p \notin \text{supp}(k)$ with matrix element $\sqrt{k_q} \geq 1$, so $a_p^\dagger a_q \psi \notin \ell^2$ for every nonzero finitely supported ψ with occupation. The interacting theory therefore requires a finite mode cutoff P_f —an ingredient of the conjecture, not an optional regularization—and any $d \rightarrow \infty$ limit demands coupling decay across modes. The free H_P , in contrast, is defined for all primes simultaneously.

The two completions are distinguished by computation:

Remark 3.12 (Numerical status of the cascade conjecture). Conjecture 3.1 has been tested by dense diagonalization at $D \leq 4096$; by shift-invert Lanczos eigen-windows ($k = 350$ interior eigenpairs placed at the median of the density of states, diagonal pivoting in the factorization, every window certified by eigenpair residuals $\leq 6 \times 10^{-8}$) up to $D = 59319$ in $d = 3$ and $D = 20736$ in $d = 4$, beyond which the LU fill exceeds the 4 GB address-space guard; and by restarted-Lanczos Krylov equilibration to $D \approx 10^5$, with restart steps sized by a Lanczos-estimated spectral radius (validated against exact eigendecomposition propagation to 10^{-12} , state-vector norm drift $\leq 10^{-13}$). The kinetic family is examined at matched effective dose $VK^2 \approx 36$, the occupation-weighted coupling, so that the dose is fixed while the truncation widens; code, seeds, parameter grids, and all result files accompany the paper (repository `MIKEAA2020/prime-numbers-in-physics`). The evidence: (a) *Level statistics*: generic couplings give GOE ratio statistics $\langle r \rangle = 0.45\text{--}0.58$ at every scale computed from $D = 625$ to 5.9×10^4 (the upper end of the band is the $d = 4$, $K = 11$, $D = 20736$ window; the sub-GOE values sit at the smallest matched-dose grids, $K \leq 14$, where the fixed effective dose leaves spectral mixing incomplete); the weak-coupling control collapses the participation ratio to $\text{PR}/D \approx 10^{-4}$ with σ_{ETH} inflated by an order of magnitude, and its $\langle r \rangle$ is *window-placement-dependent in a way the generic regime is not*: a σ -quantile sweep of the same weak configuration ($d = 3$, $K = 28$, $D = 24389$) at quantiles $\{0.02, 0.10, 0.15, 0.25, 0.5, 0.75, 0.9, 0.98\}$ gives $0.42\text{--}0.56$, non-monotone

(the dip at the 15% quantile, 0.436, is reproduced exactly in an independent repetition—density-of-states cluster structure, not solver noise), while the generic coupling at the same platform is window-dependent too, with half the amplitude: the full eight-quantile sweep of the same generic configuration ($g_0 = 1.5$, $d = 3$, $K = 28$, $D = 24389$) gives 0.464–0.540, the compression sitting at the band edges (0.464 at the 98% quantile)—an earlier three-interior-quantile check (0.505–0.514 at 0.05/0.5/0.95) sampled only the flat middle of that band; weak median value is additionally direction-dependent (0.53 at $d = 3$ versus 0.44 at $d = 4$, $D = 20736$). Any protocol claim about weak-coupling statistics must therefore pin the quantile and the direction; the robust weak-coupling diagnostic is PR/D , not $\langle r \rangle$.

(b) *ETH fails for the bare H_{tot} structurally:* H_P is a separable ramp, H_W is a per-axis adjacency (Wannier–Stark chains; propagation along the p -axis is Stark-suppressed when $g_p \lesssim \log p$), and H_{hop} is quadratic—no quartic term exists anywhere in H_{tot} , so the system is essentially single-particle on the exponent lattice $\mathbb{Z}^d$. The scaled data make this quantitative: the normalized eigenstate fluctuation $\sigma_{\text{ETH}}/\text{std}(a_j)$ of local occupations sits at 0.94–0.99—fluctuations as large as the observable’s full spread—flat in D at every coupling and every fixed dose (item (e) below); and the time-averaged occupation retains 6–24% of the initial mode’s occupation ($|\text{diag} - \text{micro}|/K$) at every scale from $D = 10^4$ to 10^5 , with no downward trend.

(c) *The diagonal quartic completion does not rescue thermalization at any accessible scale:* the dose–response is monotone localization ($\langle r \rangle$: 0.514 at $U = 0$, 0.502 at $U = 0.5$, 0.387 at $U = 2$, all at $D \approx 2 \times 10^4$; PR/D : $0.18 \rightarrow 0.03 \rightarrow 0.001$; frozen-mode memory $|\text{diag} - \text{micro}|/K \approx 0.8$).

(d) *The kinetic (density-assisted hopping) completion*—a one-term addition to the Hamiltonian, Hermitian and exactly $\hat{N}_{\text{tot}}$ -preserving, validated against a first-principles dense assembly to 10^{-15} —is the only completion that relaxes. At matched dose $VK^2 \approx 36$ on the $d = 4$, $K = 11$, $D = 20736$ platform, the occupation $\hat{n}_d$ falls from its initial value 11 to ≈ 3.5 within $\tau \approx 20$ and then drifts on a slow secular tail; the eigenstates stay delocalized with GOE statistics ($\text{PR}/D = 0.14$, $\langle r \rangle = 0.51$) where the quartic completion collapses to 0.001–0.03. The equilibration is a dose phenomenon, not a monotone onset: at $V = 0.1$ the plateau sits *away* from microcanonical ($|\text{diag} - \text{micro}|/K = 0.18$, the bare value; the short-horizon averages at the same platform are 0.10 bare and 0.81 quartic), and strong dose localizes, as the occupation-weighted matrix elements predict (effective coupling VK^2 : at $V = 1$, $\langle r \rangle \rightarrow 0.41$ and $\text{PR}/D \rightarrow 0.05$; at fixed $V = 0.3$ on the $d = 3$, $K = 28$ grid, where $VK^2 \approx 235$, $\langle r \rangle = 0.36$ with $\text{PR}/D = 0.016$).

(e) *Scaling at matched dose; the diagonal ensemble.* Two legs decide the conjecture. The eigenstate fluctuation ratio $\sigma_{\text{ETH}}/\text{std}(a_j)$ is flat: 0.95–0.98 along the $d = 3$ ladder $D = 3375 \rightarrow 59319$ (a factor 17.6 in D) and 0.95–0.97 along the $d = 4$ ladder $D = 4096 \rightarrow 20736$, with exact full-spectrum values 0.68–0.83 at $D \leq 10^4$; the ratio compares the fluctuation to the window’s own spread, which contains the smooth energy dependence, so it is window-width-dependent and not by itself a scaling diagnostic—item (h) develops the absolute normalization, in which the fluctuation decays with exponent 0.19 ($d = 3$) and 0.22 ($d = 4$) against the thermal $1/2$, receding from the strong-ETH scaling. The diagonal ensemble $\langle a \rangle_{\text{diag}} = \sum_n |\langle n | \psi_0 \rangle|^2 \langle n | a \rangle$ is computed exactly at $D \leq 10^4$: there the finite-time plateau equals it to $|\text{plateau} - \text{diag}|/K \leq 0.005$ (the dephasing identity $\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \langle a(t) \rangle dt = \langle a \rangle_{\text{diag}}$ holds with no ETH input, and nested late-time window averages supply the convergence diagnostic). But at matched dose the diagonal ensemble is not microcanonical: $|\langle a \rangle_{\text{diag}} - \langle a \rangle_{\text{micro}}|/K = 0.040 \rightarrow 0.061 \rightarrow 0.092$ along $d = 4$ ($D = 4096 \rightarrow 20736$) and ≈ 0.05 (stable in absolute value, ≈ 0.87) along $d = 3$ ($D = 3375 \rightarrow 9261$), with no closing trend. The $d = 4$, $D = 20736$ trajectory was extended to $\tau = 300$ in checkpointed wall-clock chunks (a chunk boundary is an ordinary Lanczos restart; the resumed trajectory agrees with an independent step-size choice to 2×10^{-4} in the observable): its nested late-time averages saturate at $\langle \hat{n}_d \rangle_{\text{diag}} \approx 4.40 \pm 0.03$, while the microcanonical value is 3.35 and the early transient ($\tau \leq 22$) passes through 3.39, close to it. A finite-horizon coincidence with the microcanonical value is therefore not evidence of thermalization; leg (ii) is a statement about the $D \rightarrow \infty$ limit of the diagonal ensemble, and at accessible scales that limit shows no convergence (Figure 2). (The

kinetic couplings destroy the diagonal dominance on which default partial pivoting relies in the shift-invert factorization; diagonal pivoting (`diag_pivot_thresh` = 0) restores tractability—the generic $d = 3$, $K = 28$ factorization drops from 257s to 3s—with every eigen-window certified by eigenpair residuals $\leq 6 \times 10^{-8}$ and a pivoting cross-check at $D = 14641$ agreeing to 0.002 in $\langle r \rangle$.) An $E \rightarrow \infty$ statement cannot be falsified by any finite- D simulation; a *scaling* criterion can be—and on the present evidence every scaling diagnostic is flat. (f) *Label-scrambled controls: what is arithmetic and what is graph-generic*. The null hypothesis “the diagonal’s arithmetic arrangement is irrelevant once the density of states and the graph are fixed” is tested directly: permute the on-site energies $\sum_p k_p \hbar \omega_0 \log p$ among the vertices of the *same* graph (same couplings, same adjacency and hop amplitudes, same eigenwindow seed), which preserves the exact multiset of diagonal energies and every off-diagonal matrix element and destroys only the step structure—the fixed increment $\hbar \omega_0 \log p$ per occupation quantum along each axis, incommensurate across axes. Four window pairs at $D = 20736$ – 24389 decide (Figure 1). At matched kinetic dose the level statistics are *scramble-insensitive* ($\langle r \rangle$ 0.518 $\rightarrow$ 0.512 at $V = 0.3$; 0.509 $\rightarrow$ 0.502 at matched dose): the GOE band of item (a) is a property of the graph, not of the primes. The generic ($V = 0$) family is not: scrambling moves $\langle r \rangle$ from 0.581 (above GOE) to 0.510 at $d = 4$, $K = 11$, and collapses the participation ratio by $2\times$ there and by $12\times$ at $d = 3$, $K = 28$ (PR/ D 0.182 $\rightarrow$ 0.015, σ_{ETH} 1.8 $\rightarrow$ 4.5). The arithmetic diagonal is thus a *transport organizer*: its fixed incommensurate steps support resonance manifolds—near-degenerate site families produced by multiplicative near-relations among the primes—that a random assignment of the same energies does not. What the diagonal’s arithmetic structure buys, at these scales, is delocalization and eigenstate smoothness, not level repulsion. (g) *Finite-size scaling and asymptotic distance*. Along the matched-dose ladders the deviation $\langle r \rangle - r_{\text{GOE}}$ runs from -0.082 ($D = 4096$; -0.076 at $D = 3375$) to -0.002 ($D = 19683$) with scatter, and linear fits in $1/\log D$ have large residuals: the computed range neither confirms nor excludes the GOE limit (Figure 1(a)). The distance to the asymptotic regime is itself fixed and reportable: the largest truncation computed ($d = 4$, $K = 17$) has $E/\hbar \omega_0 \leq 17 \log 210 \approx 91$ and typical prime-factor counts near $\log n \approx 91$ of ≈ 4.5 , whereas the Erdős–Kac statistics of the cascade require $\log \log x \rightarrow \infty$: every finite- D diagnostic in this paper is a statement about a pre-asymptotic window. (h) *Eigenstate fluctuation scaling in absolute units*. Normalize the same fluctuation by the a priori spread of the local occupation over the $(K+1)^d$ product basis, $\text{std}_{\text{basis}} = \sqrt{K(K+2)/12}$: $\kappa = \sigma_{\text{ETH}}/\text{std}_{\text{basis}}$. Along the matched-dose ladders κ is $O(1)$ and decays slowly: 0.50 $\rightarrow$ 0.29 at $d = 3$ ($D = 3375 \rightarrow 59319$), a fitted power law $\kappa \propto D^{-0.185 \pm 0.021}$ ($n = 9$, $R^2 = 0.92$), and $D^{-0.222 \pm 0.003}$ at $d = 4$ on the three *LU* points ($D = 4096 \rightarrow 20736$), extended to a four-point fit $D^{-0.255}$ by the factorization-free tier at $D = 3.84 \times 10^4$ (item (i); $\kappa = 0.250$, 276 certified pairs); exact full-spectrum diagonalizations reproduce the window protocol to $\leq 11\%$ and give consistent fixed-fraction exponents (-0.10 at $d = 3$, $D \leq 9261$; -0.21 at $d = 4$, $D \leq 10^4$). Two calibrated benchmarks locate the decay (Figure 3). Strong ETH predicts $\kappa \propto D^{-1/2}$ with the prefactor fixed by the shell statistics of the observable: the exact fixed-fraction benchmark B_{frac} (a 5% quantile shell at the median) sits a factor $4.6 \rightarrow 11.3$ ($d = 3$) and $4.6 \rightarrow 7.2$ ($d = 4$) below the measured κ , and the distance grows along both ladders—the fluctuation recedes from the strong-ETH scaling as the truncation widens. The protocol-matched benchmark B_{count} (the basis shell spanned by the eigen-window itself, $N_{\text{shell}} \approx 10^3$ states) is approached more slowly (κ/B_{count} falls $15.0 \rightarrow 6.4$ at $d = 3$); in effective-dimension terms the eigenstates behave as random combinations of $N_{\text{eff}} = 2\sigma_{A,\text{shell}}^2/\sigma_{\text{ETH}}^2 \approx 8 \rightarrow 23$ states, growing roughly as $D^{0.3}$ —two orders of magnitude below both the shell and the dimension. The structure behind the number is visible directly: within a single median-energy window the eigenstate values $\langle E|\hat{n}_1|E \rangle$ span $[0.3K, 0.94K]$, and $\kappa\sqrt{\text{PR}}$ (the fluctuation against the support-uniform level $\sqrt{2}$) grows from 8.5 to 28 along the $d = 3$ ladder: the eigenstate weight is correlated with the occupation it measures—the same slow mode that produces the secular drift of item (e). The decay itself is organized by the arithmetic diagonal: label-scrambled controls at matched dose hold κ flat

at 0.43–0.46 while their participation ratio *falls* (PR/D : $0.082 \rightarrow 0.035$ over the same ladder). The level statistics show no size trend (slope of $\langle r \rangle$ in $\log D$: 0.013 ± 0.020 ; mean 0.500). At the scales computed, the strong-ETH leg of C4 therefore carries quantitative negative evidence: the fluctuation decays two and a half times more slowly than the thermal scaling and its distance from the fixed-fraction benchmark grows by a factor ≈ 2.5 ; a pre-asymptotic crossover at larger D cannot be excluded (item (g)), but no trend toward it is visible in the computed range.

(i) *A factorization-free window tier, and where its own ceiling sits.* The window diagnostic is bounded by LU fill; the same $k = 350$ median-energy eigenpairs can be computed without any factorization by block Chebyshev subspace iteration on $B = (H - c)^2$, where c is the density-of-states median (the shift-invert target: the smallest eigenvalues of B are the eigenvalues of H nearest c). A low-pass Chebyshev $T_M(\xi(B))$, $\xi(b) = 1 + 2(b_w - b)/(b_{\text{high}} - b_w)$, amplifies the passband $\pm t$ (holding the block, sized by stochastic Lanczos quadrature with 8–12 probes of 360 fully reorthogonalized Lanczos steps—the block itself cannot see past its own boundary) against a stopband bounded by 1; Rayleigh–Ritz in float64 follows every sweep, filtering runs in float32 with a float64 polish, and every selected pair is certified by its residual. Validated against the LU tier at four platforms: the eigenvalue sets agree to machine precision where both run ($\max |\Delta\lambda| \leq 4 \times 10^{-11}$ at $D = 9.3 \times 10^3$ and 10^4 ; κ within 0.6% at $D = 6.9 \times 10^3$), and at the near-ceiling platform $D = 2.4 \times 10^4$ the tier certified 289 of 350 pairs within its sweep budget with κ within 2.4% (the certified subset is central, and its bias is measured, not assumed). Applied to the first LU -infeasible $d = 4$ platform ($K = 13$, $D = 3.84 \times 10^4$, degree $M = 3400$), it certifies 276 pairs at residuals $\leq 1.8 \times 10^{-6}$ with GOE statistics ($\langle r \rangle = 0.511$) and gives $\kappa = 0.250$ (the subset bias of 2–3% raises this to ≈ 0.256): the $d = 4$ ladder now reads $0.449 \rightarrow 0.367 \rightarrow 0.313 \rightarrow 0.250$ over $D = 4096 \rightarrow 38416$, a four-point fit $\kappa \propto D^{-0.255}$ against the thermal $D^{-1/2}$, with the distance from the fixed-fraction benchmark still growing (κ/B_{frac} : $7.2 \rightarrow 7.8$) past the LU ceiling. The tier’s own ceiling is quantitative: its cost is governed by the Chebyshev degree $M \approx \ln(\varepsilon) R/(2t)$, and on this family R is *dose-pinned*—the fully occupied corners of the exponent lattice carry row sums proportional to $VK^2(d - 1)$, so $R \approx 570$ at $VK^2 = 36$ in $d = 4$ against a physical band of ≈ 70 —while $t \propto D^{-1}$ at fixed protocol. The degree requirement therefore grows linearly in D along the ladder: the LU -fill ceiling at $D = 2.1 \times 10^4$ is replaced by a compute-degree ceiling roughly one rung higher (hours per configuration at this platform), and the estimator, its passband calibration, and its certification machinery are in place for larger-scale resources.

(j) *The alternative factorization-free routes, measured; the upper-edge tier to $D \approx 10^5$.* Three further factorization-free routes for the median window were implemented and measured on this family, and all stall below the Chebyshev tier’s reach: restarted Lanczos (ARPACK, which=’LA’) on the folded operator $-(H - c)^2$ at subspace budgets 240/460/700 converges 0 of 160 wanted pairs after 60–121 implicit-restart cycles (the window boundary is a spectral continuum, and single-vector implicitly restarted Lanczos cannot separate the cluster edge without solves; a first invocation with which=’LM’ silently returned the *opposite* spectral end—far-edge eigenpairs, which certify like any others—caught only by a window-position check); LOBPCG preconditioned by a doubled incomplete LU of the shifted system overflows on the indefinite factorizations; unpreconditioned LOBPCG contracts at ≈ 0.96 per iteration; and a soft Gaussian-filter estimator needs polynomial degree $\sim \|H\|/t \approx 5 \times 10^3$ at this band. What *does* converge at every scale without any factorization is the *upper-edge* window: $\text{eigsh}(H, \text{which=’LM’})$ returns the $k = 350$ largest-eigenvalue pairs at residuals $\leq 1.3 \times 10^{-11}$ (40 s at $D = 20736$, 233 s at $D = 104976$), and the same window diagnostics (30 bins) give a second fluctuation ladder: the *ratio* $\sigma_{\text{ETH}}/\text{std}$ is flat at 0.871–0.895 along $d = 4$ ($D = 4096 \rightarrow 104976$, a factor 25.6) and 0.835–0.839 along $d = 3$ ($D = 3375 \rightarrow 59319$), while in the absolute units of item (h) the edge κ falls on the same slow track as the median. The edge is a different sector— $\langle r \rangle_{\text{edge}} = 0.38\text{--}0.43$ (Poisson-like), $\text{PR}/D = 0.054 \rightarrow 0.009$ (semi-localized band-edge chains)—so the edge ladder is reported as a corroborating measurement of the fluctuation ratio’s flatness past both ceilings, not as a substitute for the median window.

The bin-width control (eleven to one hundred sixteen levels per bin) moves the ratio by less than 0.05 at every grid where it is computable. (k) *Matched-dose trajectories at the three largest grids.* The resumable Krylov trajectories extend the equilibration record to the grids past the LU ceiling: at $D = 38416$ ($K = 13$) the running average climbs through the microcanonical value on the same slow secular tail as item (e) (3.96 at $\tau = 20$, 4.21 at 39, 4.43 at 59, 4.62 at 78, 4.79 at 96; microcanonical 3.956), and at $D = 65536$ and $D = 104976$ the trajectories reach $\tau = 44$ and 36 in wall-clock chunk sessions (~ 2.2 h across twelve resumable chunks)—the dephasing horizon shrinks with the level spacing, and the diagonal-ensemble saturation of item (e) is beyond day-scale compute at these dimensions; the records and checkpoints accompany the pilots. (l) *The sector falsifier ladders and the classical limits.* The occupation sectors of Proposition 3.10 are one-parameter thermodynamic ladders in their own right (dimension $n = \binom{S+d-1}{d-1}$, no truncation active), and the strong-ETH question sharpens there into a joint decision object: the fluctuation ratio κ of the mode-1 occupation in absolute units $\sigma_{\text{ETH}}/\sqrt{S(S+2)/12}$, the ratio statistic $\langle r \rangle$, and the Haar-in-shell benchmark. The two-mode sector is closed by the solvability of Proposition 3.10(ii): at matched dose $VS^2 = 36$ the measured κ converges to the analytic floor $1/30 = 0.0333$ ($\kappa = 0.0333$ at $S = 8192$) with $\langle r \rangle = 1$ at every scale, and at fixed $V = 0.3$ the floor decays as $1/S$ ($\kappa = 1.0 \times 10^{-5}$ at $S = 2048$) while the picket statistic persists—an obstruction by exact solvability, with the fluctuation ratio and the level statistic in overt disagreement, so neither diagnostic alone decides. The U -legs localize ($\kappa \rightarrow 0.9999$, $\text{PR}/n \rightarrow 5 \times 10^{-4}$ at $U = 2$, matched dose). In $d = 3$ and $d = 4$ the ladders separate by convention: at matched dose, κ decays from 0.30 to 0.043 ($d = 3$, $n = 231 \rightarrow 2.0 \times 10^4$) with $\langle r \rangle \rightarrow 0.386\text{--}0.396$ (Poisson—the free-reduction limit), and in $d = 4$ it turns around and rises (0.115 at $n = 4495$ to 0.182 at $n = 2.3 \times 10^4$) with $\langle r \rangle$ in the 0.47–0.53 band; at fixed $V = 0.3$, $d = 3$ sits on a plateau $\kappa \approx 0.196$ at $n = 1.3 \times 10^4$ with clustered statistics $\langle r \rangle = 0.148$, and $d = 4$ decays slowly ($\kappa \sim n^{-0.22}$, $\langle r \rangle \rightarrow 0.40\text{--}0.42$), all with delocalized eigenstates ($\text{PR}/n = 0.12\text{--}0.25$) and κ a factor 1.4–20 above the Haar benchmark at the largest dimensions. A $\pm 30\%$ anisotropic control (V_{pq} per pair, one realization) leaves both signatures at the same level (κ plateau 0.28 in $d = 3$, 0.15 in $d = 4$), so the permutation covariance of Proposition 3.10(iii) is not the operative obstruction. The classical limits of the sector problem (the coherent-state flow on $\mathbb{C}P^{d-1}$, integrated with a validated fourth-order scheme and Benettin exponents) corroborate the split: the one-body flow of the matched-dose limit is exactly the linear flow $\alpha \mapsto e^{-iMt}\alpha$ ($\lambda_{\max} \leq 10^{-9}$, integrable at every d), while the kinetic flow of the fixed- V limit is chaotic at $d \geq 3$ ($\lambda_{\max} = 0.03\text{--}0.11$ across initial conditions at $d = 3$; 0.116–0.137 uniformly at $d = 4$; 0.003–0.004, the integrator floor, at $d = 2$). No tested convention exhibits the joint thermalization signature ($\langle r \rangle \rightarrow \text{GOE}$ with κ on the Haar line) at the accessible sector dimensions: classical chaos in the interaction-dominated limit coexists with non-thermal sector eigenstates, so the quantum-ergodicity route is open classically but not realized at the computed scales. Figure ?? shows the four panels (fluctuation ladders with benchmarks, level statistics, classical exponents, and the corner-state Krylov growth, whose $d = 2$ sequence is the binomial $\beta_n = C\sqrt{n(S+1-n)}$ exactly, as the solvability requires).

Falsifiers and test routes are listed in Section 10.

3.5 The Janus state

A natural boundary condition is the vacuum: the claim that “at the past hypothesis the phase $1^{-i\omega_0\tau} = 1$: time does not exist at $n = 1$; time turns on when the universe transitions to composite states.” Under H_P *nothing* ever transitions (Proposition 3.1), so the claim would be empty; under H_{tot} the vacuum $|1\rangle$ is *not* an eigenstate (because $H_W|1\rangle = \sum_p g_p|p\rangle$), and the statement becomes substantive and true:

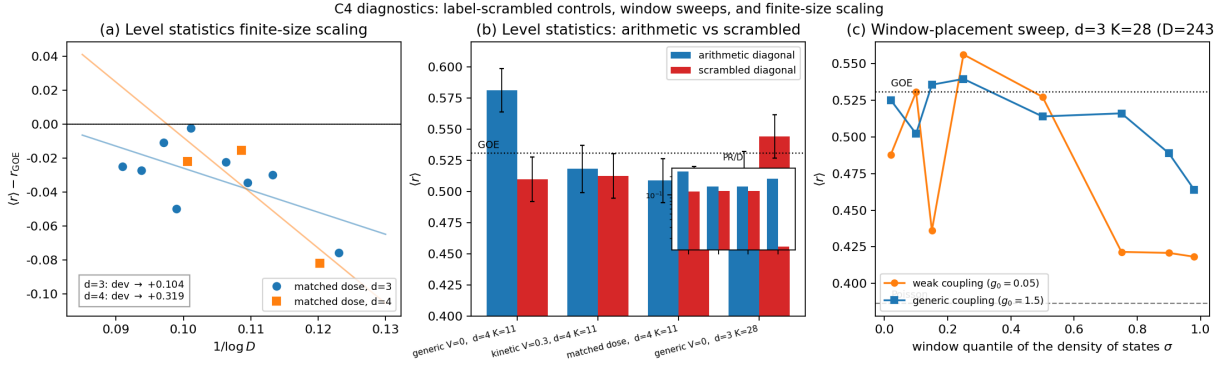


Figure 1: Audit-response controls for the C4/C5 numerics (Remark 3.12(f)–(g)). (a) Finite-size scaling of the level-statistics deviation $\langle r \rangle - r_{\text{GOE}}$ against $1/\log D$ along the matched-dose ladders ($d = 3$ circles, $d = 4$ squares; fits as lines): deviations shrink with D but the computed range does not settle the limiting value. (b) Arithmetic versus label-scrambled diagonal at four platforms ($\langle r \rangle$ with standard errors; inset: PR/D on a log scale): matched-dose $\langle r \rangle$ is scramble-insensitive; the generic family is not, and the participation ratio collapses under scrambling. (c) Window-placement sweeps at $d = 3$, $K = 28$: the weak-coupling band (0.42–0.56) and the generic band (0.46–0.54) are both quantile-dependent; the generic compression sits at the band edges.

Proposition 3.13 (Nucleation from the Janus state). *For H_{tot} with $H_W \neq 0$ and initial state $\psi(0) = |1\rangle$:*

$$\psi(\tau) = |1\rangle - \frac{i\tau}{\hbar} \sum_{p \in P_f} g_p |p\rangle + O(\tau^2), \quad (30)$$

so the total primon number grows initially at rate $d\langle \hat{N}_{\text{tot}} \rangle / d\tau = 2 \text{Im} \langle \psi | H_W | \psi \rangle / \hbar + O(\tau) = \frac{2\tau}{\hbar^2} \sum_p g_p^2 + O(\tau^2) > 0$. The unique zero-entropy state $|1\rangle$ is a regular point of the dynamics, not a fixed point; under H_P alone it is a fixed point (Proposition 3.1).

Proof. First-order time-dependent perturbation theory around the eigenstate $|1\rangle$ of H_P (the Dyson series in $H_W + H_{\text{hop}}$; H_{hop} annihilates the vacuum). $\square$

Time, in this framework, is switched on by the interaction sector, with a mechanism that exists. The stronger claims that the actual history of the universe *begins* at $|1\rangle$, and that the entropy profile is two-sided (Janus-shaped) about that event, are boundary-condition statements and are demoted to Conjectures C7 and C8; what remains as theorem is only the uniqueness and zero entropy of the floor (Theorem 2.4 and Section 6).

4 Recurrence, Periodicity, and the Fate of Eternal Return

The anti-cyclical verdict—that prime arithmetic rigorously forbids exact cyclical time and the universe cannot repeat—decomposes into a theorem, a correct application, and two qualifications. This section separates them. The resulting statement is *stronger* than the slogan, because it identifies exactly which recurrence phenomena are forbidden, which are unavoidable, and what the boundary between them depends on.

4.1 Linear independence of the prime logarithms

Lemma 4.1 (Fundamental-theorem independence). *Every finite $\mathbb{Q}$ -linear relation among the $\log p$ is trivial: the set $\{\log p : p \in \text{Pr}\}$ is $\mathbb{Q}$ -linearly independent in the finite-support sense. In particular, for distinct primes p, q , the ratio $\log p / \log q$ is irrational.*

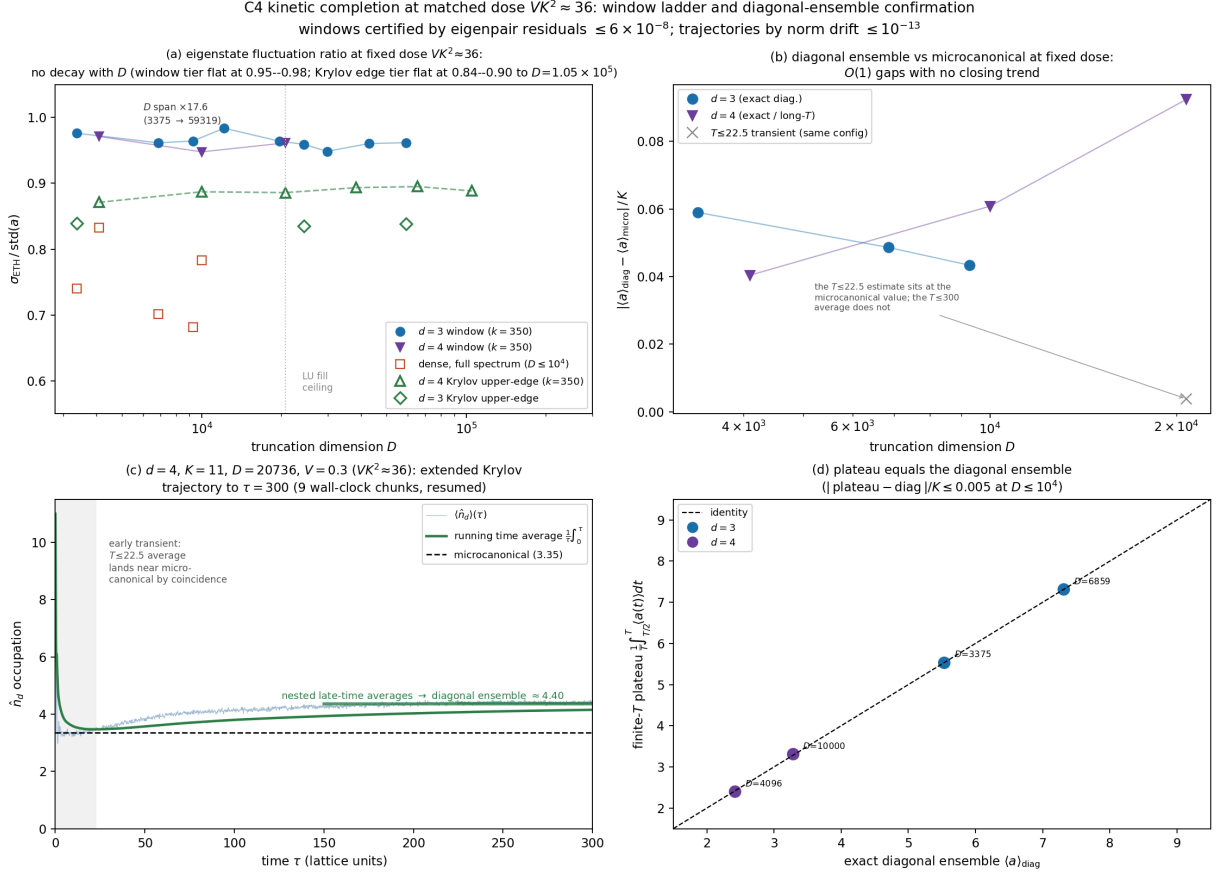


Figure 2: Matched-dose ($VK^2 \approx 36$) numerics for the kinetic completion (Remark 3.12(d)–(e)). (a) Eigenstate fluctuation ratio $\sigma_{\text{ETH}}/\text{std}(a)$ versus truncation dimension: window values (filled) and exact full-spectrum values (open); no decay across a factor 17.6 in D . (b) Distance of the diagonal ensemble from the microcanonical value, normalized by K : $O(1)$ gaps with no closing trend; the $T \leq 22.5$ estimate (cross) is a transient that coincides with the microcanonical value. (c) The $d = 4, K = 11, D = 20736, V = 0.3$ trajectory extended to $\tau = 300$: observable, running time average, nested late-time averages (diagonal-ensemble estimate ≈ 4.40), microcanonical line, and the early transient. (d) Finite- T plateau versus exact diagonal ensemble at $D \leq 10^4$ (identity line).

Proof. Suppose $\sum_{i=1}^r q_i \log p_i = 0$ with $q_i = a_i/b_i \in \mathbb{Q}$ and distinct primes p_i . Let $B = \text{lcm}(b_i)$ and $c_i = Bq_i \in \mathbb{Z}$. Then $0 = \sum_i c_i \log p_i = \log \prod_i p_i^{c_i}$ (negative exponents handled by clearing signs into the other side), so $\prod_i p_i^{c_i} = 1$. Unique factorization forces $c_i = 0$ for all i , hence $q_i = 0$. $\square$

4.2 Exact periodicity is forbidden

Theorem 4.2 (No exact return for generic states). *Let $\psi = \sum_n c_n |n\rangle$ and let $S(\psi) = \{n : c_n \neq 0\}$ be its support. The following are equivalent:*

1. *There exists $\tau > 0$ with $e^{-i\tau H_P/\hbar} \psi = \psi$;*
2. *There exists $T > 0$ such that $\omega_0 T \log n \in 2\pi\mathbb{Z}$ for every $n \in S(\psi)$;*
3. *The set $\{\log n : n \in S(\psi)\}$ is contained in $\alpha\mathbb{Z}$ for some $\alpha > 0$ (all supports are integer multiples of a common logarithmic unit; equivalently all $n \in S(\psi)$ are powers of a single integer).*

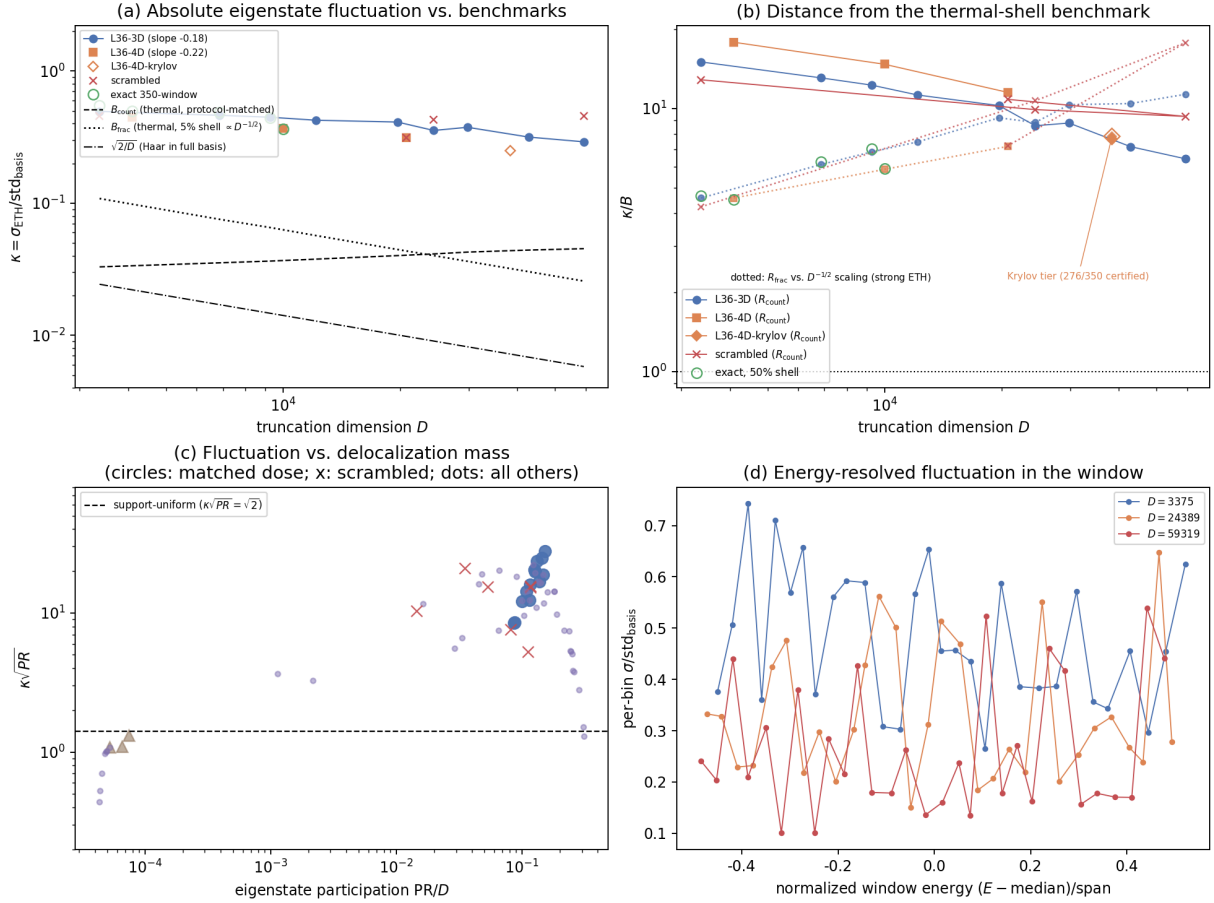


Figure 3: Absolute eigenstate-fluctuation scaling at matched dose (Remark 3.12(h)–(i)). (a) $\kappa = \sigma_{\text{ETH}}/\text{std}_{\text{basis}}$ versus truncation dimension (window values, exact 350-state selections, label-scrambled controls, and the factorization-free Krylov-tier point at $D = 3.84 \times 10^4$, open diamond: 276 of 350 pairs certified) against the calibrated benchmarks: the protocol-matched shell benchmark B_{count} , the fixed-fraction $B_{\text{frac}} \propto D^{-1/2}$ (the strong-ETH scaling), and the Haar-in-full-basis level $\sqrt{2/D}$. (b) Distance from the benchmarks: κ/B_{count} (solid) falls slowly; κ/B_{frac} (dotted) grows. (c) $\kappa\sqrt{\text{PR}}$ versus eigenstate participation across all computed window configurations (matched dose: circles; scrambled: crosses; all others: dots), against the support-uniform level $\sqrt{2}$: the fluctuation sits 10–30 times above it and grows with D . (d) Energy-resolved within-window fluctuation at three scales.

In particular, if $S(\psi)$ contains any two integers m, n that are not powers of a common integer—for example any pair among $\{2, 3\}$, $\{6, 10\}$, $\{2, 6\}$ —then ψ is aperiodic: $e^{-i\tau H_P/\hbar}\psi \neq \psi$ for every $\tau > 0$. The same holds for H_{tot} whenever its spectrum is simple and contains two levels whose difference is irrational in units of $\hbar/\omega_0$, which holds generically.

Proof. $e^{-i\tau H_P/\hbar}\psi = \sum_n c_n e^{-i\omega_0\tau \log n} |n\rangle$ equals ψ iff $e^{-i\omega_0\tau \log n} = 1$ for all $n \in S(\psi)$, i.e. iff $\omega_0\tau \log n \in 2\pi\mathbb{Z}$ for all such n . Setting $\alpha = 2\pi/(\omega_0\tau)$, this says every $\log n$ in the support lies in $\alpha\mathbb{Z}$. If $m, n \in S$ and $\log m, \log n \in \alpha\mathbb{Z}$ with $\log m/\log n \notin \mathbb{Q}$, the condition fails for every τ ; by Lemma 4.1, $\log 2/\log 3 \notin \mathbb{Q}$ etc. $\square$

This is the rigorous core of the anti-cyclic verdict: *eternal return in the exact sense requires the support of the universal state to lie in a single geometric progression of integers*, and no physically motivated state does. The ray version (periodicity up to a global phase, i.e. for physical states rather than vectors) relaxes the condition to: all ratios n/m with $n, m \in S(\psi)$ are powers of a single rational—the support lies in one multiplicative coset $\{c a^j\}$. This too fails

for any support with two multiplicatively independent ratios (e.g. $S = \{6, 12, 18\}$: the ratios 2 and $\frac{3}{2}$ are independent by unique factorization), so the aperiodicity verdict is unchanged at the level of rays.

4.3 What is *not* forbidden: three qualifications

Corollary 4.3 (Truncated systems recur). *Fix a finite truncation: primes $p \leq X$ and occupations $k_p \leq K$, so the Hilbert space is finite-dimensional and the spectrum $\{E_j\}$ of H_P (or of any self-adjoint extension) is finite. Then the evolution $U(\tau) = e^{-i\tau H/\hbar}$ is a quasi-periodic rotation, and for every state ψ and every $\varepsilon > 0$ the set $\{\tau > 0 : \|U(\tau)\psi - \psi\| < \varepsilon\}$ is relatively dense (there exists $L = L(\varepsilon, \psi)$ such that every interval of length L contains such a τ). In particular Poincaré recurrence holds in its strongest (uniform, Bohr almost-periodic) form. This is the finite-dimensional special case of Theorem 4.4 below, recorded separately because it is the form in which recurrence appears in any computation.*

Proof. On a finite-dimensional space $U(\tau) = e^{-i\tau\Lambda}$ with $\Lambda = \text{diag}(E_1, \dots, E_d)/\hbar$; the map $\tau \mapsto (e^{-i\tau E_1/\hbar}, \dots, e^{-i\tau E_d/\hbar})$ is a continuous one-parameter subgroup of the compact torus $\mathbb{T}^d$, and every orbit of a compact group rotation is uniformly recurrent: the return times to any neighborhood of the identity are syndetic. This is Kronecker's theorem in its group form [16]. $\square$

Theorem 4.4 (Uniform recurrence of the free flow, infinite system). *For every state $\psi \in \mathcal{H}_A$ and every $\varepsilon > 0$, the set of ε -return times*

$$\left\{ \tau > 0 : \|e^{-i\tau H_P/\hbar}\psi - \psi\| < \varepsilon \right\} \quad (31)$$

is relatively dense in $\mathbb{R}$. The free arithmetic flow is Bohr almost periodic for all states, with no finite invariant measure required and no truncation assumed. What fails is not ε -recurrence but its uniformity in the state: the return-time scale $L(\varepsilon, \psi)$ is unbounded as a function of ψ (it grows with the Diophantine complexity of the finite supports approximating ψ), and exact return remains forbidden by Theorem 4.2.

Proof. Fix $\varepsilon > 0$ and $\psi = \sum_n c_n |n\rangle \in \ell^2$. Choose a finite support set $F \subset \mathbb{N}_{\geq 1}$ with $\sum_{n \notin F} |c_n|^2 < \varepsilon^2/4$. The truncated flow $\tau \mapsto (e^{-i\omega_0 \tau \log n})_{n \in F}$ is a continuous one-parameter subgroup $h: \mathbb{R} \rightarrow \mathbb{T}^F$; let $H = \overline{h(\mathbb{R})}$, a compact connected subgroup of the torus. The $\mathbb{R}$ -orbit of the identity under left translation by $h(\tau)$ is dense in H by construction, and a dense orbit of an equicontinuous (group) action on a compact space is minimal; minimal equicontinuous flows are uniformly recurrent, so $T(\delta) = \{\tau : d(h(\tau), e_H) < \delta\}$ is relatively dense for every $\delta > 0$. Taking δ small enough that all phases on F deviate by less than $\varepsilon/2$ in the ℓ^2 sense (finitely many constraints), we obtain $\|U(\tau)\psi - \psi\| \leq \varepsilon/2 + \varepsilon/2$ for every $\tau \in T(\delta)$, a relatively dense set. The non-uniformity in ψ follows because F must grow as the tails of ψ shrink, and the Diophantine return times on F are unbounded in $|F|$; the quantitative form is Proposition 4.5 below. $\square$

Proposition 4.5 (Quantitative return-time bound). *Let $\psi = \sum_n c_n |n\rangle$ be supported on a finite set $F \subset \mathbb{N}_{\geq 1}$, let $m_F = \#\{p : p \mid n \text{ for some } n \in F\}$ be the number of distinct prime frequencies active in F , and let $K_F = \max_{n \in F} \max_p k_p(n)$. Then for every $\varepsilon > 0$ there exists an ε -return time*

$$\tau \leq \left(\frac{2\pi C_F}{\varepsilon} \right)^{m_F} \quad \text{with} \quad C_F := 2 + K_F m_F, \quad (32)$$

and the set of ε -return times is relatively dense with gaps of the same order (Bohr's theorem [17]).

Proof. Dirichlet’s pigeonhole on the m_F -torus: with $N \geq 2$, the $N^{m_F} + 1$ points $\{k\alpha \bmod 2\pi\}_{k=0}^{N^{m_F}}$, where $\alpha = (\omega_0 \log p)_{p|F}$, fall N^{m_F} -to-one into the $(2\pi/N)$ -grid cells of $\mathbb{T}^{m_F}$; two of them share a cell, and their difference $q \leq N^{m_F}$ satisfies $|q\omega_0 \log p|_{\bmod 2\pi} < 2\pi/N$ for every $p \mid F$. Each phase of $n \in F$ is the integer combination $\sum_p k_p(n) \omega_0 \log p$ with $\sum_p k_p(n) \leq K_F m_F$, so every phase on F deviates by less than $2\pi K_F m_F / N \leq \varepsilon$ once $N \geq 2\pi K_F m_F / \varepsilon$ (choose also $N \geq 2$ so that the wrapped sum stays in $(-\pi, \pi]$). Then $\|e^{-i\tau H_P/\hbar} \psi - \psi\|^2 \leq \sum_n |c_n|^2 |e^{i\theta_n} - 1|^2 \leq \max_n |\theta_n|^2 \leq \varepsilon^2$ with $\tau = q \leq N^{m_F}$. Relative density with gaps of the same order is the syndeticity statement of Bohr almost-periodicity [17]. $\square$

Proposition 4.5 is the quantitative content of “Diophantine slowness”, and it separates two recurrence scales that the framework needs kept apart.

Arithmetic return. A typical integer near x carries $\sim \log \log x$ distinct prime factors (Theorem 3.8), so a superposition of M integers near x has $m_F \lesssim M \log \log x$, and (32) gives

$$\tau_\varepsilon \lesssim (C/\varepsilon)^{M \log \log x} = (\log x)^{M \log(C/\varepsilon)} = (S/k_B)^{O(1)}, \quad (33)$$

where $S(E) = k_B \log x = k_B E / \hbar \omega_0$ is the microcanonical entropy: for states of fixed or slowly growing support, arithmetic ε -returns occur on times *polynomial in the entropy*. (The exponent $M \log(C/\varepsilon)$ is set by the support size and the resolution; the occupations K_F enter only the constant C_F , not the exponent.)

Generic return. A quantum system with D resolved incommensurate frequencies requires, by the same pigeonhole bound, $\tau_\varepsilon \gtrsim (\varepsilon^{-1})^D$. At energy E the arithmetic Hilbert space has $D = N(E) = e^{S/k_B}$ levels; a *generic* system on that many resolved frequencies needs $\tau_\varepsilon \gtrsim e^{e^{S/k_B} \log(1/\varepsilon)}$: *exponential in the entropy* (doubly so). The primon gas occupies the extreme short-return end of this spectrum, and the mechanism is exactly arithmetic: unique factorization compresses D level frequencies into the $m_F \ll D$ integer combinations of $\log p$.

For Conjecture C2 this converts the qualitative cutoff language into a scaling falsifier: recurrence phenomenology whose times grow faster than every power of the entropy at fixed resolution ε is inconsistent with every effective arithmetic cutoff (upper bound (33) changes only polynomially in S when the cutoff X or K is enlarged), while recurrence times polynomial in S are the arithmetic prediction. No cutoff-consistent adjustment absorbs an exponential-in- S recurrence scale; conversely, observed ε -returns at polynomial-in- S times, for states of bounded support, would be a signature of frequency compression of exactly the multiplicative kind.

The third qualification concerns the *classical* multiplicative walk, and it sharpens the boundary of the no-cycle verdict in an unexpected direction.

Proposition 4.6 (Pólya transience of the multiplicative walk). *Consider the classical random walk on the divisibility graph of Definition 3.2 restricted to $d = |P_f|$ active primes: from $k = (k_1, \dots, k_d) \in \mathbb{Z}_{\geq 0}^d$ step to $k + e_j$ with probability $1/(2d)$ and to $k - e_j$ (when $k_j > 0$) with probability $1/(2d)$, staying otherwise. Then the walk is recurrent for $d \leq 2$ and transient for $d \geq 3$ [18]. In particular for $d \geq 3$ the walk returns to the vacuum $k = 0$ (the Janus state) only finitely many times, almost surely.*

Proposition 4.6 has a striking interpretive reading: whether the arithmetic universe “cycles back to its origin” depends on the *number of active prime modes*. With one or two primes, the multiplicative universe is recurrent—a rigorous echo of cyclical cosmology. With three or more, it is transient, and the Janus point is a genuinely singular origin. The absolute prohibition thus becomes a structural, checkable dichotomy.

4.4 The verdict on cyclical time

Assembling the three legs: exact eternal return is impossible for states whose support spans incommensurable integers (Theorem 4.2); ε -approximate recurrence is universal for the *free* flow,

at every state, with return times relatively dense but astronomically non-uniform (Theorem 4.4, and Corollary 4.3 for the trivial finite case); and the *cascading* (or classical walk) dynamics is transient for $d \geq 3$ modes (Proposition 4.6), with the interacting quantum system conjecturally non-recurrent on every physical scale (Conjecture C4). The philosophical consequences are drawn in Section 9; the essential point is that the framework’s anti-cyclical verdict survives as a *layered* statement—exact return forbidden as a genuine theorem, practical return driven out of reach by Diophantine slowness, transience of the cascade for three or more active modes—rather than as an absolute metaphysical prohibition. The free arithmetic sector, left to itself, is the *most* recurrent object in the theory (Bohr almost periodicity for all states, Theorem 4.4); all irreversibility is therefore manufactured by the interaction sector and the boundary condition, never by the spectrum. This is the sharpest available form of the intuition that “the river never circles back,” and it is also the form most useful to the arrow-of-time analysis of Sections 5 and 9.

5 The Arithmetic Clock and the Problem of Time

5.1 The problem of time, stated precisely

The reconciliation problem for the two classical concepts of time is standard: in quantum mechanics, t is an external parameter and the Schrödinger equation evolves states with respect to it; in general relativity, elapsed time is the proper time along a worldline, $\tau_\gamma = c^{-1} \int_\gamma \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$, a geometric quantity that varies from trajectory to trajectory, while the canonical Hamiltonian of gravity is a constraint, $\hat{\mathcal{H}}\Psi = 0$, with no external evolution at all. The resolution is relational: time is a correlation between subsystems, realized quantum mechanically by the Page–Wootters mechanism [19], with the primon spectrum supplying the clock. This section turns that resolution into theorems, corrects one sign-of-life issue (the clock kernel never decorrelates, so the arrow of time cannot live in the clock), and quantifies the claim that “at the Janus point time does not exist.”

The Page–Wootters construction is standard: for a total state Ψ satisfying a constraint $(H_C + H_S)\Psi = 0$ of clock-plus-system, the conditional state of the system when the clock reads τ is $\psi_S(\tau) = \langle \tau | \Psi \rangle$, and for an ideal clock family this satisfies an effective Schrödinger equation in τ . All content therefore concentrates in the clock family. The analysis below supplies the *kinematical* level of this program: the clock family, the conditional-state map, and the correlation kernel. The constraint equation itself is not solved here—solving it requires the interaction sector of Section 3 and belongs to the content of C4—and no claim is made that this section dissolves the problem of time; what is proved is the exact correlation structure of the arithmetic clock, and the exact statement of what that structure forbids.

5.2 The zeta clock

Definition 5.1 (Zeta clock family). For a dimensionless regulator $\nu > \frac{1}{2}$ define the normalized clock states

$$|\tau; \nu\rangle = \zeta(2\nu)^{-1/2} \sum_{n=1}^{\infty} n^{-\nu - i\omega_0 \tau} |n\rangle, \quad (34)$$

a unit-norm family in $\mathcal{H}_A$ (unit norm because $\sum_n n^{-2\nu} = \zeta(2\nu)$ converges for $2\nu > 1$). The regulator is a resolution parameter, not a temperature: the weights $n^{-\nu}$ confine the clock to energies of order $\hbar\omega_0/(2\nu - 1)$, as computed in Proposition 5.4. A clock at regulator ν has the same mean energy as a canonical primon state at $\sigma = 2\nu$; the clock family therefore exists on *both* sides of the Hagedorn scale— $\nu \downarrow \frac{1}{2}$ is the high-energy (canonical-divergence) limit, $\nu \rightarrow \infty$ the vacuum limit—and it is the analytic object through which energies above T_H , where no Gibbs state exists, are nevertheless resolved (Section 2).

Proposition 5.2 (Clock covariance and two-time kernel). 1. (Covariance.) $e^{-i\tau' H_P/\hbar}|\tau; \nu\rangle = |\tau + \tau'; \nu\rangle$: clock translations are generated by H_P itself, so the clock reading τ is exactly the phase parameter conjugate to $\log \hat{N}$,

$$e^{-i\omega_0 \tau \log \hat{N}}|n\rangle = n^{-i\omega_0 \tau}|n\rangle. \quad (35)$$

2. (Two-time kernel.) For all τ, τ' ,

$$\langle \tau'; \nu | \tau; \nu \rangle = \frac{\zeta(2\nu - i\omega_0(\tau - \tau'))}{\zeta(2\nu)}. \quad (36)$$

3. (Positivity.) The kernel (36) is positive definite on $\mathbb{R}$: it is the Fourier transform of the probability measure $\mu_\nu = \zeta(2\nu)^{-1} \sum_n n^{-2\nu} \delta_{\log n}$ on the energy-per-mode variable $u = \log n$.

4. (Product of phases.) For a superposition $\sum_n c_n |n\rangle$ evolved for time τ , the phase acquired is $n^{-i\omega_0 \tau}$, and by unique factorization

$$n^{-i\omega_0 \tau} = \prod_p p^{-i\omega_0 \tau k_p} : \quad (37)$$

each prime contributes an elementary phase generator, and composite evolution is the product of prime evolutions. This is the exact sense in which “primes provide the elementary generators of spectral phase order.”

Proof. (1) is immediate term by term. (2) $\langle \tau'; \nu | \tau; \nu \rangle = \zeta(2\nu)^{-1} \sum_n n^{-2\nu} e^{i\omega_0(\tau - \tau') \log n} = \zeta(2\nu)^{-1} \zeta(2\nu - i\omega_0(\tau - \tau'))$, since $e^{i\omega_0 \Delta \tau \log n} = n^{i\omega_0 \Delta \tau}$. (3) The kernel $K(\Delta \tau) = \int e^{i\omega_0 \Delta \tau u} d\mu_\nu(u)$ is the characteristic function of μ_ν composed with a dilation, hence positive definite by Bochner’s theorem. (4) follows from unique factorization applied to the logarithm of the phase. $\square$

Equation (36) is the framework’s signature formula: *the distinguishability of two readings of the arithmetic clock is governed by the Riemann zeta function*. What follows adds the facts that make it usable.

Theorem 5.3 (Almost-periodicity of the clock: the arrow cannot live here). For every $\nu > \frac{1}{2}$, the kernel $K(\Delta \tau) = \zeta(2\nu - i\omega_0 \Delta \tau) / \zeta(2\nu)$ is a Bohr almost-periodic function of $\Delta \tau$ [17] (the Dirichlet series converges absolutely on the line $\sigma = 2\nu > 1$). Consequently:

1. $|K|$ does not decay along any sequence: $\limsup_{|\Delta \tau| \rightarrow \infty} |K(\Delta \tau)| = 1$.
2. For every $\varepsilon > 0$ the set $\{\Delta \tau : |K(\Delta \tau) - 1| < \varepsilon\}$ is relatively dense: clock correlations resynchronize infinitely often, in both time directions, at every resolution ν .

No limit $\nu \downarrow \frac{1}{2}$ escapes this: at every fixed ν the kernel is two-sided symmetric, and the singular limit only narrows the correlation peak (Proposition 5.4(2)) without ever selecting a direction. The high-energy clock is more resolved, not directed.

Proof. On the line $\sigma = 2\nu > 1$, $\zeta(s) = \sum_n n^{-s}$ is an absolutely convergent Dirichlet series with $\sum_n n^{-\sigma} < \infty$, hence a (uniform limit of finite exponential sums) Bohr almost-periodic function of the imaginary part [17]. Almost-periodic functions have relatively dense ε -almost-periods, and $K(0) = 1$. $\square$

This theorem resolves a hidden tension in the framework. The zeta object appears twice: as the clock (“the distinguishability of two times is governed by the zeta function”) and as the engine of the thermodynamic arrow (“the cascade up the mountain of composite numbers”). Theorem 5.3 shows these two roles cannot be fused: an almost-periodic clock is *recurrent* and

carries no arrow. In the framework the arrow is therefore relocated entirely to the interaction sector and the boundary condition (Conjecture C4 and C7), while the clock contributes the neutral, arrow-free, maximally stable phase reference. This is a structural clarification, not a loss: it says that *within the framework, time measurement and time's arrow are functionally separated objects*, exactly as relational philosophy of time requires.

5.3 Energy–time duality of the arithmetic clock

Proposition 5.4 (Resolution duality). *For the clock family $|\tau; \nu\rangle$:*

1. *The mean energy is $\langle H_P \rangle_\nu = \hbar\omega_0(-\zeta'/\zeta)(2\nu) = \frac{\hbar\omega_0}{2\nu-1} + O(\hbar\omega_0)$ as $\nu \downarrow \frac{1}{2}$.*
2. *The correlation peak is asymptotically Lorentzian: as $\nu \downarrow \frac{1}{2}$, uniformly for bounded $\omega_0\Delta\tau/(2\nu-1)$,*

$$|K(\Delta\tau)|^2 = \frac{(2\nu-1)^2}{(2\nu-1)^2 + \omega_0^2\Delta\tau^2} (1 + o(1)), \quad (38)$$

with half-width $\Delta\tau_{1/2} \asymp (2\nu-1)/\omega_0$.

3. *Consequently $\langle H_P \rangle_\nu \Delta\tau_{1/2} \rightarrow \hbar$ as $\nu \downarrow \frac{1}{2}$: the zeta clock asymptotically saturates the Mandelstam–Tamm energy–time limit. Time resolution is bought one-for-one with clock energy, and the clock never becomes “cheap.”*
4. *Conversely, as $\nu \rightarrow \infty$ the state concentrates on $|1\rangle$: $|\tau; \nu\rangle \rightarrow |1\rangle$, energy $\rightarrow 0$, and $K \rightarrow 1$ for all $\Delta\tau$: the clock resolves nothing. At the Janus limit the clock degenerates—the quantified form of the claim that “at $n=1$ time does not exist.”*

Proof. (1) is Theorem 2.6 at $\sigma = 2\nu \downarrow 1$. Positivity of $\langle H_P \rangle$ requires $\nu > 1/2$, consistent with Definition 5.1. (2) Near the pole, $\zeta(\sigma - it) = (\sigma - 1 - it)^{-1} + O(1)$, so $|K|^2 = |\sigma - 1|^2 / ((\sigma - 1)^2 + t^2) + o(1)$ with $t = \omega_0\Delta\tau$ and $\sigma - 1 = 2\nu - 1$. (3) Multiply (1) and (2). (4) $n^{-2\nu}$ concentrates on $n = 1$; $K(t) = \zeta(2\nu - it)/\zeta(2\nu) \rightarrow 1$ since $\zeta(2\nu - it) \rightarrow 1$ and $\zeta(2\nu) \rightarrow 1$. $\square$

5.4 Reconciliation of the three times

With the clock machinery in place, the tripartite reconciliation can be stated as a compact conditional statement. *Quantum-mechanical time* is the reading of a relational clock in the classical-clock limit: it is recovered when the clock subsystem is large, weakly coupled, and quasi-classical, in which case $\tau_{\text{clock}} \approx t$ for all practical observers. *Relativistic time* is the proper-time functional along worldlines: it is the geometric content of the clock's embedding, determining which phase increments accumulate along which trajectory. *Quantum-gravitational time* is the constraint-generated phase correlation: with the primon clock, τ is conjugate to $\log \hat{N}$, and evolution is the multiplicative phase flow $n^{-i\omega_0\tau}$. The three are not rivals but levels: the block-level correlation structure ($\hat{\mathcal{H}}\Psi = 0$), its geometric projection (proper time), and its semiclassical limit (external t). All incompatibilities of principle dissolve at the kinematical level; what remains open is whether the *specific* identification of the clock with the primon spectrum is realized (Conjecture C1), and how the geometric sector emerges from the arithmetic one (Conjectures C3, C9).

The tripartite reconciliation is compactly stated:

Relativity provides the geometry; quantum mechanics provides the spectrum; the primes provide the arithmetic order of the spectrum; and time is the phase relation among prime-ordered states measured along geometric worldlines.

The measurement clause is Proposition 5.2(1) (clock covariance); the ordering clause is unique factorization, i.e. Proposition 5.2(4); the arrow clause is conspicuously absent, by Theorem 5.3, and lives in the interaction sector.

6 Thermodynamics, Gravity, and the Holographic Bridge

6.1 The exact counting law

A repeatedly used statement is “the number of states below energy E grows like $e^{E/\hbar\omega_0}$.” The exact statement is a one-line identity, and its precision is what makes the holographic bridge in Section 6.2 clean.

Theorem 6.1 (Log-count identity). *Let $N(E) = \#\{n \in \mathbb{N}_{\geq 1} : \hbar\omega_0 \log n \leq E\} = \lfloor e^{E/\hbar\omega_0} \rfloor$, and write $x = E/\hbar\omega_0$ and $r(E) = \log N(E) - x$. Then:*

1. (Global envelope.) For all $E \geq 0$: $-\log 2 \leq r(E) \leq 0$.
2. (Sharp two-sided bound, $E \geq \hbar\omega_0 \log 2$.)

$$\log(1 - e^{-x}) \leq r(E) \leq 0, \quad (39)$$

with both bounds sharp: $r \uparrow 0$ as E crosses a level from above, and $r \downarrow \log(1 - e^{-x})$ as E approaches a level from below.

3. (Exponential decay.) $|r(E)| \leq -\log(1 - e^{-x}) \leq 2e^{-x}$: the defect converges to zero exponentially. The counting function itself remains a staircase forever: N jumps by one at every level $E = \hbar\omega_0 \log m$, and the jump of $\log N$ there is $\log(1 + \frac{1}{m}) \sim \frac{1}{m}$ —the granularity persists at every scale while its amplitude decays like e^{-x} .
4. (Smoothing.) For every probability kernel w supported in $[-\Delta, \Delta]$ with Δ fixed, the smoothed count $\tilde{N} = N * w$ satisfies, uniformly in $E \geq \hbar\omega_0(\log 2 + \Delta)$,

$$\log \tilde{N}(E) = x + \log \int e^t w(t) dt + O(e^{-x}) : \quad (40)$$

the growth rate of every fixed-window smoothing of the entropy is exactly $1/\hbar\omega_0$ up to exponentially small corrections, at every energy scale.

In particular the microcanonical entropy $S(E) = k_B \log N(E)$ satisfies

$$S(E) = \frac{E}{T_H} + \delta(E), \quad |\delta(E)| \leq k_B \log \frac{1}{1 - e^{-E/\hbar\omega_0}} \leq 2k_B e^{-E/\hbar\omega_0} \quad (E \geq \hbar\omega_0 \log 2), \quad (41)$$

with $T_H = \hbar\omega_0/k_B$. (The unsmoothed entropy is a staircase; its pointwise microcanonical temperature is undefined between levels. The content of the Hagedorn reading is item (iv): the growth rate is energy-independent, so energy input converts into state multiplicity at a fixed exchange rate.)

Proof. $N(E) = \lfloor e^x \rfloor$ because $\hbar\omega_0 \log n \leq E \iff n \leq e^x$. Write $y = e^x$ and $\{y\} = y - \lfloor y \rfloor \in [0, 1)$, so $r = \log(1 - \{y\}/y) \in [\log(1 - 1/y), 0]$ since $\lfloor y \rfloor \geq y - 1$ and $\{y\} < 1$. The lower bound is approached as $\{y\} \uparrow 1$ (E approaching a level from below), the upper bound attained as $\{y\} \downarrow 0$. Decay: $-\log(1 - u) \leq 2u$ for $u \leq \frac{1}{2}$, and $\{y\}/y < 1/y = e^{-x} \leq \frac{1}{2}$ for $x \geq \log 2$. Jumps: on $y \in [m, m+1)$, $\log N = \log m$ and the jump at $y = m+1$ is $\log((m+1)/m) \sim 1/m$. Smoothing: $N(E+t) = e^{x+t} - \{e^{x+t}\}$, so $\tilde{N}(E) = e^x \int e^t w(t) dt - R(E)$ with $0 \leq R(E) = \int w(t) \{e^{x+t}\} dt \leq 1$; hence $\log \tilde{N} = x + \log \int e^t w(t) dt + \log(1 - R(E)e^{-x}/\int e^t w)$ and the last term is $O(e^{-x})$, uniformly in E . $\square$

Two consequences of the corrected defect estimate are load-bearing later, and both follow from item (iii). First, at matched entropy scale the integer-counting fluctuation is $O(k_B e^{-E/\hbar\omega_0})$: far smaller than the logarithmic corrections of black-hole entropy, so those corrections cannot originate from the floor sector at all—Section 6.2 turns this from a caveat into a falsifier (the

counting contributes no $\log A$ term; any such term must be carried by the dictionary). Second, the fluctuation sector of the gravitational counting function is carried not by the floor but by the *prime*-counting oscillations: the explicit formula $\psi(x) = x - \sum_{\rho} x^{\rho}/\rho + \dots$ has fluctuations bounded by $O(\sqrt{x} \log^2 x)$ under the Riemann hypothesis—the content of von Koch’s equivalence [20]: the Riemann hypothesis is *equivalent* to the statement that arithmetic fluctuations about the smooth classical term are as small as square-root-small. Read in the framework’s dictionary, RH becomes: *quantum fluctuations of the arithmetic sector about the classical continuum are optimally suppressed*. That is a rigorous reformulation of “RH = spectral stability”, and it is the strongest form of that claim that mathematics currently licenses.

6.2 The holographic bridge

The derivation of the Bekenstein–Hawking entropy from prime state counting is a natural target. This paper isolates exactly one nontrivial input and shows the rest is Theorem 6.1.

Conjecture 6.1 (Arithmetic holographic bound). *The maximal primon excitation energy accessible in, or associated with, a spacetime region whose boundary has area A is*

$$E_{\max}(A) = \frac{\hbar\omega_0}{4\ell_P^2} A + \hbar\omega_0 s(A), \quad s(A) = O\left(\log \frac{A}{\ell_P^2}\right), \quad (42)$$

independently of the detailed matter content, where $\ell_P = \sqrt{G\hbar/c^3}$ is the Planck length. The term $\hbar\omega_0 s(A)$ is the dictionary error term: it is the only slot through which logarithmic corrections can enter, since the counting theorem contributes none (Theorem 6.1(iii)).

Theorem 6.2 (Conditional Bekenstein–Hawking). *Assume Conjecture 6.1. Then the arithmetic entropy of the region is*

$$S(A) = k_B \log N(E_{\max}(A)) = \frac{k_B A}{4\ell_P^2} + k_B s(A) + \delta(E_{\max}(A)), \quad |\delta| \leq 2k_B e^{-E_{\max}/\hbar\omega_0}, \quad (43)$$

which is the Bekenstein–Hawking entropy [21, 22] with the area coefficient exact and every logarithmic correction carried by the dictionary term $s(A)$ and none by the counting. The pure model ($s = O(1)$) predicts no $-\alpha \log(A/\ell_P^2)$ term at all; a computed correction of that class (several ensembles give $-\frac{3}{2} \log A$) therefore falsifies the pure form and measures s . Moreover, for a Schwarzschild black hole of mass M , $A = 16\pi G^2 M^2/c^4$ and $A/4\ell_P^2 = 4\pi G M^2/(\hbar c)$; the choice $E_{\max} = Mc^2$ and $\hbar\omega_0 = 2k_B T_H^{\text{BH}}$ normalizes (42) exactly at fixed horizon, with $T_H^{\text{BH}} = \hbar c^3/(8\pi G M k_B)$ the Hawking temperature.

Proof. Substitute (42) into Theorem 6.1. For the Schwarzschild case, $E_{\max}(A)/\hbar\omega_0 = A/4\ell_P^2$ by (42); with $E_{\max} = Mc^2$ this forces $\hbar\omega_0 = 4Mc^2\ell_P^2/A = 4Mc^2 \cdot (G\hbar/c^3)/(16\pi G^2 M^2/c^4) = \hbar c^3/(4\pi G M) = 2k_B T_H^{\text{BH}}$. $\square$

Two remarks delimit the bridge; both are registered under C3.

Remark 6.3 (The factor-of-two ambiguity). The Hawking identification above is exact only per-horizon: since $T_H^{\text{BH}} \propto M^{-1}$, a *universal* ω_0 cannot equal $2T_H^{\text{BH}}$ for all masses. Either ω_0 is universal and the identification holds to within $O(1)$ horizon-dependent factors, or the arithmetic scale is set per horizon. Which alternative survives is a genuine open question that the conjecture form makes explicit; a presentation that silently treats ω_0 as universal and the identification as exact would hide it. In both alternatives the *area coefficient* is exact, because the counting defect is exponentially small (Theorem 6.1(iii)); all residual freedom lives in the dictionary term $s(A)$.

Remark 6.4 (What the bridge does and does not derive). The bridge converts an area law into an *energy* cutoff statement plus an exact counting theorem; it does not derive the cutoff. This is precisely the logical position of the holographic principle in mainstream theory, translated into arithmetic terms. The conversion loss is provably zero in the area coefficient (Theorem 6.1 has no hidden degeneracy factors, since the primon spectrum is simple, Theorem 2.4), and every logarithmic correction is imported through the dictionary term $s(A)$: any slack in Conjecture 6.1 is slack in the physics, not in the counting.

6.3 Singularities and the Hagedorn wall

The singularity claims swing between two poles: one places the primon gas *at* the singularity (the reported conformal system near the black-hole center), later reading declares that singularities *do not exist*. The reconciliation is a standard move of effective field theory, and it is worth stating once, cleanly, because it removes the apparent contradiction:

Remark 6.5 (Resolution of the singularity tension). “Singularity” denotes the boundary of validity of the continuum effective description, not a locus in the fundamental theory. In the reconstructed framework: the geometric (Einstein) description fails where its implicit continuum counting diverges; the arithmetic description remains finite at all energies (Theorem 2.4 has no finite accumulation point, and all quantities are finite below T_H). The pole of ζ at $s = 1$ is the exact analytic signature of the breakdown: it is a *critical* (Hagedorn) divergence, not a curvature divergence. What the Big Bang and black-hole interiors “are” in this framework is the regime where the geometric effective theory must be replaced by the arithmetic one; the primon gas is the UV-complete description of *that regime*, which is compatible with it also being located, in the geometric coordinates of the effective theory, “near the singularity.” Both readings are thereby retained, each in its proper tense. The associated physical conjecture (C6) is that the Hagedorn wall T_H replaces the singular point: approach to the singularity is approach to a critical temperature at which energy converts into primon creation rather than curvature growth.

6.4 Emergent gravity and the hierarchy problem

The resolution of the hierarchy problem—gravity as an entropic, $1/N$ -suppressed collective effect [23–25]—is the most speculative of its physics claims, and it is demoted in full. But the demoted form can be made unusually concrete, because the arithmetic sector supplies an explicit N .

Conjecture 6.2 (Arithmetic hierarchy). *The Newton constant is set by the number of arithmetic degrees of freedom below the effective cutoff E_{cut} :*

$$\frac{1}{G} \asymp \frac{\hbar}{c^3} \omega_0 N_{\text{eff}}(E_{\text{cut}}), \quad N_{\text{eff}}(E) = \lfloor e^{E/\hbar\omega_0} \rfloor, \quad (44)$$

up to a dimensionless $O(1)$ factor, so that

$$\log \frac{M_{\text{Pl}}}{M_{\text{EW}}} \asymp \frac{E_{\text{cut}}}{2\hbar\omega_0} \implies E_{\text{cut}} \asymp 2 \log\left(\frac{M_{\text{Pl}}}{M_{\text{EW}}}\right) \hbar\omega_0 \approx 74 \hbar\omega_0 \quad (45)$$

(using $M_{\text{Pl}}/M_{\text{EW}} \approx 10^{16}$).

The honest content of Conjecture 6.2 is a *relocation* of the hierarchy problem, stated with numbers: the question “why is gravity so weak?” becomes “why does the effective arithmetic universe of the electroweak regime terminate at $n \sim e^{74} \approx 10^{32}$ integers?”—which is a question about the cutoff structure of the arithmetic sector, not a fine-tuning of a Lagrangian parameter. The framework does not answer it; it converts it into a well-posed question with a single dimensionless number to explain, register entry C11. This is strictly more than a bare $1/N$ -suppression narrative with no N , and strictly less than a derivation of G .

6.5 The classical limit and the pole at unity

The classical-limit condition reads: as $s \rightarrow 1^+$, the coarse-grained arithmetic partition function should reproduce classical geometry, the pole $\zeta(s) \sim 1/(s-1)$ representing the divergent continuum limit. This paper restates this as a theorem-flanked analogy: the pole is exactly the Hagedorn divergence (Theorem 2.6); the smooth term x of $\psi(x)$ is the classical contribution and the zeros contribute the fluctuations (von Koch’s equivalence [20]); and “renormalization removing the pole” means passing to the finite part of the Laurent expansion, whose constant term is Euler’s constant γ —the arithmetic framework’s first finite counterterm. Whether the geometric sector (Einstein dynamics, $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$) can actually be *derived* as this coarse-graining is the deepest part of Conjecture C1 and is not claimed.

7 Entanglement: Arithmetic Superpositions and Holism

7.1 Composite integers are product states

The framework’s informal layer identifies quantum entanglement with multiplicative composition: “a composite number is mathematically entangled: it cannot be described independently of its prime factors”; “entanglement is not non-local communication, it is multiplicative inseparability”; “a composite number like $n = p_A \cdot p_B$ is a single, inseparable mathematical entity.” As quantum theory this is false. The correction relocates arithmetic entanglement to where it genuinely lives—superpositions across prime-mode bipartitions—and preserves the arithmetic content as structure rather than as a false identification.

Theorem 7.1 (Product structure of the occupation basis). *Let $P_A, P_B \subset \text{Pr}$ be disjoint with $P_A \cup P_B = \text{Pr}$, and let $\mathcal{F}_A, \mathcal{F}_B$ be the submonoids of $\mathcal{F}$ supported on P_A, P_B . The monoid isomorphism $\mathcal{F} \cong \mathcal{F}_A \times \mathcal{F}_B$ (unique factorization: $k \mapsto (k|_{P_A}, k|_{P_B})$) induces a unitary*

$$\mathcal{H}_A \cong \mathcal{H}_A \otimes \mathcal{H}_B, \quad |k\rangle \mapsto |k_A\rangle \otimes |k_B\rangle, \quad (46)$$

under which every occupation basis vector is a product state. In particular $|pq\rangle = |p\rangle_A \otimes |q\rangle_B$ for $p \in P_A, q \in P_B$: a composite integer is an unentangled configuration of its prime factors. Entanglement across the bipartition is a property of superpositions $\psi = \sum c_{ij}|i\rangle_A|j\rangle_B$ of rank > 1 in the coefficient tensor, exactly as in ordinary quantum mechanics, and of nothing else.

Proof. The restriction map is a bijection of free commutative monoids by unique factorization; the induced map on ℓ^2 bases is a bijection of orthonormal bases preserving inner products; extend linearly. Product structure of each basis vector is the definition of the tensor identification. $\square$

The composite-as-entangled picture thus becomes a precise dichotomy: the integer n *labels* a product configuration; *arithmetic entanglement* is superpositional. The dichotomy is stronger than the identification it replaces, because it carries quantitative content: entanglement entropies, Schmidt ranks, and Bell inequalities.

Definition 7.2 (Arithmetic entanglement). For a bipartition $\text{Pr} = P_A \sqcup P_B$ and a pure state $\psi \in \mathcal{H}_A \otimes \mathcal{H}_B$, the *arithmetic entanglement entropy* $S_{\text{ent}}(\psi)$ is the von Neumann entropy of $\rho_A = \text{Tr}_B|\psi\rangle\langle\psi|$. ψ is entangled iff $S_{\text{ent}}(\psi) > 0$, iff the coefficient tensor (c_{ij}) in the occupation basis has rank > 1 when reshaped as a matrix.

Example 7.3 (The six–thirty-six Bell pair). Take $P_A = \{2\}, P_B = \{3\}$. Then

$$\frac{1}{\sqrt{2}}(|6\rangle + |36\rangle) = \frac{1}{\sqrt{2}}(|2\rangle_A|3\rangle_B + |4\rangle_A|9\rangle_B) \quad (47)$$

is maximally entangled across the bipartition: Schmidt rank 2, equal Schmidt values $1/\sqrt{2}$, $S_{\text{ent}} = \log 2$. The superposition of the integers 6 and 36 *is* an arithmetic Bell pair. More

generally $\frac{1}{\sqrt{2}}(|ab\rangle + |cd\rangle)$ with a, c powers of distinct primes of P_A and b, d of P_B , $(a, b) \neq (c, d)$, is maximally entangled.

Proposition 7.4 (Bell violation in the arithmetic sector). *Restrict local measurements to the two-dimensional subspaces $V_A = \text{span}\{|2\rangle, |4\rangle\} \cong \mathbb{C}^2$ and $V_B = \text{span}\{|3\rangle, |9\rangle\} \cong \mathbb{C}^2$, identified with qubits. On the Bell state of Example 7.3, the standard spin settings along the xz -plane achieve the CHSH value $2\sqrt{2}$, the Tsirelson bound [26, 27]. Hence the arithmetic sector, equipped with superpositional states, exhibits exactly standard quantum non-locality of correlations; the violation is destroyed if the superposition is replaced by either basis vector, i.e. by any definite integer.*

Proof. The restriction of the state to $V_A \otimes V_B$ is the standard Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$; the CHSH computation is the textbook one. $\square$

7.2 Splitting primes as natural entanglement resources

The Gaussian sector supplies a construction the rational sector lacks, and it realizes two images at once (the “complex primon overlay” of the five-dimensional picture and the claim that symmetry breaking is prime splitting).

Proposition 7.5 (Splitting-prime EPR pairs). *Let $p \equiv 1 \pmod{4}$ be a rational prime, so that $p = \pi\bar{\pi} = N(\pi)$ in $\mathbb{Z}[i]$ with π a Gaussian prime, $\pi \sim \bar{\pi}$ conjugate and non-associate. Define the splitting state*

$$|p\rangle_{\text{split}} = \frac{1}{\sqrt{2}}(|\pi\rangle \otimes |\bar{\pi}\rangle + |\bar{\pi}\rangle \otimes |\pi\rangle) \quad (48)$$

on the two Gaussian modes generated by π and $\bar{\pi}$. Then:

1. $|p\rangle_{\text{split}}$ is maximally entangled across the $(\pi, \bar{\pi})$ bipartition ($S_{\text{ent}} = \log 2$);
2. it is invariant under the conjugation involution C (which swaps the two tensor factors), so it is even under the arithmetic charge conjugation of Section 8;
3. it is the natural correlate, inside the arithmetic Hilbert space, of the splitting of p in $\mathbb{Z}[i]$ —the phenomenon whose statistical frequency is exactly $1/2$ by Chebotarev/Dirichlet (Section 8).

Proof. (1) The two tensor factors are qubit-spaces $\text{span}\{|\pi\rangle, |\bar{\pi}\rangle\}$; the state is the symmetric Bell state. (2) C exchanges $|\pi\rangle \leftrightarrow |\bar{\pi}\rangle$ in each factor, and the state is manifestly symmetric. (3) The correspondence $p \mapsto \{\pi, \bar{\pi}\}$ is the Dedekind–Kummer splitting theorem for $\mathbb{Z}[i]$ [12]. $\square$

Proposition 7.5 is the precise version of the “phantom layer of Gaussian primes weaving through the solid prime nodes”; the weaving is entanglement between conjugate prime modes, quantitatively $\log 2$ per split integer, and it exists only for the splitting class $p \equiv 1 \pmod{4}$, which by Dirichlet’s theorem has natural density $1/2$. The inert class $p \equiv 3 \pmod{4}$ contributes *no* entanglement resource: those primes remain prime in $\mathbb{Z}[i]$. The arithmetic sector thus possesses an intrinsic, algebraically defined binary—split versus inert—with exact statistics. Section 8 builds the force-sector dictionary on it; it is also the natural habitat for the idea that the weak (chiral) interaction lives in the complex sector.

7.3 Measurement, decoherence, and records

With entanglement defined correctly, the measurement picture—“a measurement is an interaction that creates a thermodynamically irreversible, decohered record in the prime spectrum”—translates into standard quantum mechanics with an arithmetic state space, and the standard strengths *and* standard weaknesses apply.

Remark 7.6 (Measurement as record formation). In this framework a measurement is an interaction between a system and a macroscopic apparatus/environment whose net effect is to entangle the system’s superposition across the bipartition “apparatus modes versus system modes” so strongly that the reduced state of the system is, to observational accuracy, diagonal in the pointer basis (einselection) [28]. The arrow enters through the thermodynamic cost of stabilization (Landauer’s principle [29]): records can be written reliably only in the direction of increasing multiplicity of environmental states, i.e. along the cascade of Conjecture C4. Rocks measure rocks, photons measure dust grains, and consciousness is a consumer of records, not a precondition for them—each clause survives, now attached to the correct mechanism (superposition entanglement, Theorem 7.1) rather than the incorrect one (composite inseparability).

Remark 7.7 (The improper-mixture problem). Decoherence diagonalizes reduced density matrices; it does not, by itself, select one outcome. The framework elects an Everettian resolution [30], registered as interpretive (Conjecture C12) rather than theorem-grade. What the arithmetic sector adds is structural, not probabilistic: branching in $\mathcal{H}_A$ is branching of *factorization histories*—worldlines through the divisibility graph—and the Born-rule weights are the squared amplitudes of superposed integers. The 2019 reversal experiments on small qubit registers [31] are consistent with this picture in the same sense as with standard decoherence: small closed systems can be complex-conjugated back; macroscopic entangled cascades cannot, because the required inverse unitary acts on $\sim 10^{80}$ correlated modes and its probability is beyond every Poincaré time. The experiments were performed by Lesovik et al. and implemented on an IBM quantum computer.

7.4 Bell’s trilemma and the rejection of superdeterminism

Superdeterminism can be rejected with the argument that Bell correlations arise from “multiplicative inseparability” of composites. That mechanism is false (Theorem 7.1), so the rejection must rest on other grounds: the standard Everettian one, plus the arithmetic sector’s own contribution:

1. *Realism, global not local.* The arithmetic Hilbert space and the universal state are observer-independent; individual prime modes carry no pre-existing local values (the occupation basis is a basis, not a value assignment). This is exactly the quantum theory’s own position: realism preserved globally, rejected locally.
2. *Locality, no-signaling not local action.* Arithmetic-entangled states (Examples 7.3, Proposition 7.5) exhibit Bell correlations across spacelike separation (Proposition 7.4) while the reduced states are untouched by distant operations: $\rho_B = \text{Tr}_A \rho_{AB}$ is invariant under local unitaries on A . No usable signal, energy, or conserved quantity propagates superluminally; the no-signaling theorem is a theorem of the tensor structure that Theorem 7.1 now correctly exhibits.
3. *Statistical independence, preserved.* Measurement settings and hidden variables are not pre-correlated by a cosmic conspiracy. The correlations of the Bell state are *created* by the state’s preparation (a superposition), not imposed on separate local beables. Superdeterminism saves locality only by denying independence; the arithmetic framework keeps independence and gives up local realism of modes, exactly as loophole-free experiments [32] force [26, 27].

The corrected position is the standard one, reached with the arithmetic sector’s own structure. The philosophical residue—that “space” itself is emergent from entanglement structure, so that demands for local causal explanation between entangled modes are a category error—remains available as an interpretation (register C12), in the same epistemic tier as the ER=EPR-type conjectures it echoes. What is theorem-grade is narrower and cleaner: in the arithmetic sector, *non-locality of correlations is a property of superposed integers, local realism of prime modes is false, and no-signaling is exact.*

8 The Arithmetic Gauge Sector

8.1 From Lie unification to Galois unification

The framework proposes that the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ is the “macroscopic, continuous shadow” of the Absolute Galois group $G_{\mathbb{Q}} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, with the Langlands program as the dictionary, prime splitting as symmetry breaking, and Gaussian primes as the chiral (weak) sector. As physics this is speculation; as mathematics it can be anchored on all fours, because each ingredient of the analogy corresponds to an actual theorem. This section therefore proceeds in two layers: a *theorem layer* that fixes what is true about splitting statistics and symmetry structure, and a *dictionary layer* that is demoted in its entirety (Conjecture C9).

The theorem layer rests on three classical results, which we state in the form the analogy uses them.

Theorem 8.1 (Kronecker–Weber). *Every finite abelian extension $K/\mathbb{Q}$ is contained in a cyclotomic field $\mathbb{Q}(\zeta_m)$; conversely every cyclotomic field is abelian over $\mathbb{Q}$ [12]. Combined with Dirichlet’s theorem on primes in arithmetic progressions [33] and the splitting criterion*

$$p \nmid m, \text{ } p \text{ splits completely in } \mathbb{Q}(\zeta_m) \iff p \equiv 1 \pmod{m}, \quad (49)$$

the abelian sector of $G_{\mathbb{Q}}$ is completely described by residue classes of primes: the reciprocity map of class field theory is, explicitly, $p \mapsto p \pmod{m}$.

Theorem 8.2 (Dedekind–Kummer splitting). *Let $K = \mathbb{Q}(\theta)$ with θ an algebraic integer of minimal polynomial f , and p a rational prime not dividing the index $[\mathcal{O}_K : \mathbb{Z}[\theta]]$. Then the factorization pattern of f modulo p into irreducibles of degrees $d_1, \dots, d_r$ coincides with the splitting pattern of p in $\mathcal{O}_K$: $p\mathcal{O}_K = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ with residue degrees $d_1, \dots, d_r$ [12]. Symmetry breaking and prime splitting are the same operation.*

Theorem 8.3 (Chebotarev density). *Let $K/\mathbb{Q}$ be a finite Galois extension with group G , and $\sigma \in G$ any element. The set of rational primes p (unramified in K) whose Frobenius class $\text{Frob}_p \subset G$ equals the conjugacy class of σ has natural density $|\text{cl}(\sigma)|/|G|$ [34]. In particular: splitting statistics are group statistics. Dirichlet’s theorem is the abelian case (with Theorem 8.1).*

For the working example that carries the whole source dictionary, specialize to $K = \mathbb{Q}(i)$, $G = \text{Gal}(\mathbb{Q}(i)/\mathbb{Q}) = \{1, \sigma\}$ with σ the complex conjugation $a + bi \mapsto a - bi$:

Class of p	Condition	Behavior in $\mathbb{Z}[i]$	Density (Chebotarev)
Split	$p \equiv 1 \pmod{4}$	$p = \pi\bar{\pi}$, two conjugate primes	$1/2$
Inert	$p \equiv 3 \pmod{4}$	p remains prime	$1/2$
Ramified	$p = 2$	$2 = -i(1+i)^2$, one prime squared	0 (finite)

8.2 The dictionary

With the theorem layer fixed, the force-to-field dictionary can be stated so that every arithmetic entry is backed and every physical equation is demoted:

Standard Model object	Arithmetic object	Status of the pairing
U(1) gauge structure	Abelian extensions, cyclotomic fields $\mathbb{Q}(\zeta_m)$; the “photon” is the reciprocity map $p \mapsto p \pmod{m}$	Theorem layer: KW, Dirichlet (Thm 8.1); pairing demoted (C9)
Chiral $SU(2)_L$, parity violation	Quadratic/non-abelian extensions; the non-trivial automorphism σ (conjugation) acts as the arithmetic C ; split/inert dichotomy as handedness	Theorem layer: splitting table above; pairing demoted (C9, C10)
$SU(3)$ color, confinement	Inert primes ($p \equiv 3 \pmod{4}$) and their non-abelian analogues; higher-degree splitting patterns with non-commuting Frobenius classes	Theorem layer: Chebotarev (Thm 8.3); pairing demoted (C9)
Higgs mechanism, spontaneous symmetry breaking	Ramification and splitting: the unified base $\mathbb{Q}$ extends to K ; primes then <i>choose</i> splitting patterns, with exact group-valued statistics	Theorem layer: Kummer (Thm 8.2); pairing demoted (C9)
Coupling unification, charge quantization	Langlands functoriality: transfer of automorphic representations between L -groups; the dictionary itself	Mathematics: deep conjecture, proven for GL_n over function fields [35]; physics: demoted (C9)
CP violation, CKM phase	The σ -odd interaction sector: asymmetric couplings $w(\pi) \neq w(\bar{\pi})$; the non-vanishing $L(1, \chi_{-4}) = \pi/4$ as the arithmetic asymmetry functional	Theorem layer: factorization (19); model demoted (C10)

Each row now has the same structure: a true theorem about number fields, an honest conjecture about the pairing. The table is a bookkeeping of identifications, not a construction: no row supplies a gauge field, a connection, an action, a mass gap, or any confinement or scattering observable, and none is derived anywhere in this section. Each row’s physical content is exactly its conjectural statistics—restrictions, densities, hierarchies—and nothing more; the dynamical and geometric content of the Standard Model would have to be carried by constructions that this paper does not attempt. Two entries deserve expansion because they carry the most distinctive physical claims.

8.3 Arithmetic charge conjugation, parity, and time reversal

The arithmetic model of the discrete symmetries can be defined exactly on the Gaussian sector.

Definition 8.4 (Arithmetic C , P , T). On $\mathcal{H}_{\mathbb{Z}[i]}$ (Definition 2.9):

1. C is the unitary implementing $\text{Gal}(\mathbb{Q}(i)/\mathbb{Q})$: $C|\pi\rangle = |\bar{\pi}\rangle$ on Gaussian-prime modes, extended multiplicatively. It is an involution, $C^2 = I$.
2. P is any order-two relabeling of modes that plays the role of spatial inversion (e.g. a fixed-point-free involution of a finite mode set).
3. T is the antiunitary complex conjugation K in the occupation basis: it reverses all phases, $K n^{-i\omega_0\tau} K = n^{+i\omega_0\tau}$.

Proposition 8.5 (Symmetry structure of the Gaussian sector). 1. $[C, H_{\mathbb{Z}[i]}] = 0$ and $[K, H_{\mathbb{Z}[i]}] = 0$: the free Gaussian Hamiltonian is exactly C -, T -, and CPT -symmetric, because it is built from norms $N(\pi)$, which are conjugation-invariant, with real coefficients.

2. Consequently any arithmetic model of CP violation must reside in the interaction sector: couplings $w(\pi)$ with $w(\pi) \neq w(\bar{\pi})$ in the walk/hop Hamiltonians of Section 3.3. The σ -even/ σ -odd decomposition $H = H_+ + H_-$ (eigenparts under C) is an orthogonal splitting of the interaction space, and the “amount of CP violation” is the norm ratio $\|H_-\|/\|H_+\|$ together with the relative phases of the w ’s.
3. The composition $CPT = CP \cdot K$ is antiunitary; if $w(\pi) = \overline{w(\bar{\pi})}$ (the natural hermiticity condition), the full H_{tot} is CPT -symmetric, the arithmetic echo of the CPT theorem holding by construction rather than by axiomatic field theory.

Proof. (1) $N(\bar{\pi}) = N(\pi)$; diagonal operators with real eigenvalues commute with the occupation-basis conjugation K . (2) The even/odd split under an involutive unitary is standard. (3) Direct computation: CP exchanges the couplings $w(\pi) \leftrightarrow w(\bar{\pi})$; combined with K (which conjugates the phases), hermiticity in the stated form is exactly the invariance condition. $\square$

The parallel with the Standard Model is exact in structure and honest in status: in both theories, the free (kinetic) sector is CP -symmetric; the violation lives in the charged interaction sector (the CKM phase there, the asymmetric weights $w(\pi)$ here); and CPT holds by the hermiticity structure. The claim that “the weak interaction maps to the complex sector and its rare CP violation to complex phases” becomes Conjecture C10: *the CKM phase and baryon asymmetry are the arithmetic index of the σ -odd interaction sector, accumulated along the cascade*. Its falsifiable edge is sharp: the model predicts that CP -violating and T -violating observables couple to *split-versus-inert* statistics (the only binary the Gaussian sector possesses), so any derived branching asymmetry must be expressible through L -function ratios at saddles—a constraint precise enough to check and to fail.

8.4 Vaccaro’s program, translated

Vaccaro proposes that T -violation is not a curiosity of the weak sector but the microphysical origin of dynamics itself: with exactly T -symmetric laws, the universe would be a static block without localized happening [36]. The arithmetic translation is natural in this framework: the σ -odd couplings H_- are precisely what distinguishes the interacting block from the frozen block. Under H_P alone the universe is frozen (Proposition 3.1); under $H_P + H_+$ it evolves but with exactly reciprocal processes (detailed balance holds, and the walk is unbiased); under $H_P + H_+ + H_-$ there is a microscopic bias, a ratchet, and the direction of the cascade acquires a microphysical anchor in addition to the boundary condition. Conjecture C10 subsumes this: the ratchet, compounded by the exponential state growth (Theorem 2.6), is the framework’s candidate engine for baryogenesis—a “chiral arithmetic cascade” with a mechanism that exists.

8.5 What Langlands does and does not supply

Finally, the boundary of the unification layer. The Langlands program ([35] for the proven function-field GL_n case) supplies a conjectural correspondence between Galois representations and automorphic forms: in the framework’s language, between the arithmetic sector’s symmetry data and the analytic (eigenmode) data of adelic groups. The slogan—“the continuous gauge symmetries are the low-energy shadows of discrete Galois symmetries”—requires, at minimum, that such correspondences survive specialization to the finite set of splits that the Standard Model actually uses. Nothing currently known implies this; nothing currently known forbids it. This paper’s contribution is to isolate the minimal well-posed core:

Conjecture 8.1 (Galois–gauge dictionary, minimal form). *There exists an assignment from the Standard Model’s gauge and flavor data to number fields and their Galois representations such that: (i) the observed charge lattice corresponds to a lattice of splitting restrictions of the form (49); (ii) branching ratios of force sectors equal Chebotarev densities of the associated Frobenius classes; and (iii) the hierarchy of coupling strengths corresponds to the ramification hierarchy of the fields involved.*

Condition (ii) is the conjecture’s empirical hook: Chebotarev densities are rational numbers with denominator $|G|$ —the arithmetic GUT predicts that observed branching statistics, at the unification scale, organize into ratios of small integers divided by group orders. That is a checkable structural signature, quite unlike the free parameters of Lie-group GUTs, and it is the kind of prediction that would falsify C9 outright if the statistics came out transcendental or unsplittable.

The hook has been made operational as a pre-registered test protocol and applied to the Z -pole branching table. The statistic is the best class-equation-constrained dictionary fit, $T = \min \sum_i (f_i - c_i/N)^2 / \sigma_i^2$ over group orders $N \in [5, 360]$ and integers $c_i \geq 1$ with $c_i \mid N$ and $\sum_i c_i = N$ (class sizes divide the group order by orbit–stabilizer, and the classes partition the group); the minimization is exact by dynamic programming over integer partial sums. The null distribution is Monte-Carlo-calibrated on structureless (Dirichlet) tables at the measured uncertainty pattern, and the decision rule is fixed in advance of the fit: the signature is present iff the observed fit is at least as good as the null’s best 5%. Power is calibrated by injected truth tables at $N = 6, 36$, and 360 (one per decade of the registered range): the detection probability is 1.000 at the registered threshold at *current* Z -pole precision. Applied to the exhaustive five-channel Z table (hadronic, e , μ , τ , invisible; PDG values and uncertainties), the protocol returns $T = 4.1 \times 10^5$ (best fit $N = 48$, $c = (24, 2, 3, 3, 16)$) at the null’s 59th percentile: the signature is absent, and the absence is decisive rather than under-powered—each channel individually misses every candidate value c/N by at least 2.8σ (the hadronic channel by 221σ), and all uncertainties would have to inflate by a factor 208 for the best constrained fit to become acceptable. The Z -scale exclusion is consistent with the dictionary’s unification-scale placement and carries no confirmatory content; the frozen protocol (statistic, null, decision rule, power tables) applies unchanged to future precision tables, for which the statistical floor of a Z factory with $\sim 10^{12}$ recorded decays improves the current precision by a further factor $\sim 3 \times 10^{-3}$.

9 Arrows of Time and the Philosophy Register

The philosophical layer of the framework is unusually rich: it adjudicates between A-theory and B-theory, answers the vertiginous question, classifies the intrinsic and extrinsic camps, reconstructs the Past Hypothesis, rejects cyclical time, demotes causality, and rejects superdeterminism. The task of this section is different from the mathematical sections: the claims are not theorems and should never pretend to be; the task is to state each position *conditional on* the conjectural superstructure, mark its epistemic tier, and keep the ones that are well supported while discarding the arguments that leaned on loose mathematics. Every item below is either (a) a conditional consequence of registered conjectures, (b) an interpretive stance, or (c) a theorem already proved in Sections 3–8.

9.1 The extrinsic arrow, structurally grounded

The position in the intrinsic/extrinsic debate is extrinsic, but “not arbitrary”: the arrow is a structural asymmetry of the block, not a primitive flow. This section retains this and sharpens it into a three-source decomposition:

1. *Geometric order* (relativistic causal structure): a time-orientable spacetime supplies an earlier/later order and light cones—the riverbed, not the flow. This is standard general relativity and carries no arrow.
2. *Statistical multiplicity* (the arithmetic Hagedorn structure): Theorem 6.1 fixes $S(E) = E/T_H + O(1)$; the state space grows exponentially in one direction of complexity. This is theorem-grade but is an asymmetry of *counting*, not yet of *time*: it becomes a temporal arrow only under a boundary condition that places low multiplicity in one temporal direction.
3. *The boundary condition* (the Past Hypothesis, arithmetic form): the low-entropy past is the Janus state $|1\rangle$ or its neighborhood (unique zero-entropy state, Theorem 2.4). That the universe’s actual history occupies such a neighborhood at one end is Conjecture C7, and the two-sided (Janus-point) version—entropy increasing away from $|1\rangle$ in both temporal directions—is the arithmetic realization of the Carroll–Chen and Barbour et al. constructions [37–39], registered as C8.

The arrow formula (“causal order + time orientation + low-entropy boundary + spectral state counting”) survives verbatim, but its ingredients are sorted by tier: the first is geometry, the second is a theorem, the third is a conjecture. Nothing requires time to flow; everything requires the block to be asymmetric.

9.2 The hierarchy of arrows

The hierarchy—an arithmetic master arrow from which thermodynamic, causal, and psychological arrows derive, with the cosmological arrow as boundary condition—survives with the master arrow’s engine supplied. The version adopted here:

Arrow	In this framework	Status
Arithmetic (master)	Growth of available multiplicity $N(E)$; cascade of Conjecture C4	Conjecture (engine) + Theorem (counting)
Thermodynamic	$S = k_B \log \Omega$ along the cascade; second law as overwhelming-multiplicity drift	Conditional on C4, C7
Causal	Record formation: asymmetric decoherence structure [28, 29]; “ c causes e ” = e contains a thermodynamically stable record of c	Conditional; derivative, not fundamental (cf. [40, 41])
Psychological	Memory as record-bearing physical process; indexical “now”	Conditional; interpretive tier
Cosmological	Expansion as the geometric enabler of the arithmetic cascade; alignment of arrows contingent on epoch	Conditional; observation-compatible
Weak (CP/T)	σ -odd interaction sector (Proposition 8.5); micro-ratchet, not master arrow	Conjecture C10

Two notes attach to this table. First, the causal arrow’s derivation presupposes the thermodynamic arrow, so causal theories of time inherit the boundary-condition problem rather than

solving it. Second, the weak arrow is *demoted*: nothing in the mathematics forces CP violation to be a separate fundamental arrow, but nothing derives it from the counting either; it enters as an independent interaction-sector fact (Conjecture C10), consistent with Vaccaro’s program [36] without being implied by it.

9.3 Cyclical time: the verdict

Section 4 proves the layered verdict: exact eternal return is impossible for generic states (Theorem 4.2); ε -return is nevertheless universal for the free flow, for every state, with relatively dense but astronomically non-uniform return times (Theorem 4.4); and the multiplicative cascade is transient for $d \geq 3$ modes (Proposition 4.6). The philosophical consequence—“the river can never flow in a circle”—survives in the layered form: exact cycling is a theorem-grade impossibility, practical cycling is excluded by Diophantine slowness of the returns, and the cascading dynamics genuinely never returns for three or more active modes. The framework thereby gains two things: the recognition that the *free* spectrum is maximally recurrent, so that every arrow of time must be manufactured by the interaction sector; and a *parameter-controlled* transition between recurrent and transient arithmetic universes (the number of active prime modes), which converts a metaphysical prohibition into a physically meaningful question.

9.4 Eternalism, indexicality, and the vertiginous question

The position—block universe at the fundamental level, perspectival observers inside it, rejection of both the moving spotlight and egocentric presentism—is retained, with its argument tightened at the two points where it leaned on loose physics.

Remark 9.1 (Indexical perspectival realism). The “democracy of perspectives” does not follow from the symmetry of the constraint $\hat{\mathcal{H}}\Psi = 0$ alone. A constraint’s symmetry does not by itself create perspectives; what creates them is the decomposition of Ψ into entangled clock–system–observer branches, which is exactly the Page–Wootters structure of Section 5 with the corrected entanglement of Section 7. Each perspective is an indexical fact—“this branch, this clock reading”—of the Kaplanian kind: “now” and “I” are token-reflexives, and their referents are fixed by the utterance event, not by a metaphysical spotlight [42]. Hellie’s vertiginous question (“why is this perspective the one occurring?”) [7] is thereby answered in the same move that answers “why am I here?” on a map: the question has a referent-fixing answer and no metaphysical remainder; what remains is the phenomenology, which is real but not ontology. Conitzer’s strengthened A-theory arguments [9] and Hare’s egocentric presentism [8] are then answered with the felt flow as the experience of record formation along the cascade—but now the cascade has a mechanism that exists (Section 3) and the flow’s substrate is the superpositional entanglement structure, not composite inseparability.

Remark 9.2 (What would make the block falsifiable as an interpretation). The eternalist tier of the framework is interpretation, but it is *tethered* interpretation: it inherits its standing from the constraint formalism (C1), the Page–Wootters clock (theorems), and the cascade conjecture (C4). If C4 failed—if the interacting arithmetic dynamics failed to thermalize—the psychological arrow would lose its engine and the perspectival story would need replacement. This tethering answers the rhetorical question “is the framework well-developed enough to confidently pick a side?”: yes, conditional on the register; no, unconditionally.

9.5 Past Hypothesis as arithmetic necessity: theorem and conjecture

The most striking cosmological claim here is that the low-entropy past is not a brute fact but the arithmetic floor: “the universe began at $n = 1$ because 1 is the only number with no prime factors.” Its correct and incorrect parts split cleanly. *Correct, and theorem-grade*: $|1\rangle$ is the unique state of zero energy and zero entropy (Theorem 2.4); it is the unique minimum

of complexity; any trajectory through the multiplicative lattice that touches the floor must pass through it; and a state exactly at $|1\rangle$ is stationary (with the interaction sector switched off, Proposition 3.1)—the quantitative “time does not exist at the floor” (Proposition 5.4(4)). *Conjectural, and demoted*: that the actual universal state was ever at or near $|1\rangle$ (C7), and that the entropy profile is two-sided about it (C8). What the demotion buys is precision about what a derivation would need: a mechanism that places the initial condition at the floor without fine-tuning. The arithmetic framework’s candidate answer—the floor is the *only* state at which the thermodynamic and complexity orderings degenerate, so any boundary-condition principle that is symmetry-like must select it—is registered as the motivation for C7, in the same tier as Wallace’s quantum-cosmological reading of the Past Hypothesis [5] and Price’s symmetry principles [6]. The claim that the framework “dissolves” the Past Hypothesis is downgraded to: the framework *localizes* the Past Hypothesis to a single, mathematically distinguished state, which is progress, not dissolution.

9.6 The Big Crunch, Gold universes, and reverse-arrow regions

The claim under test is that the thermodynamic arrow cannot reverse in a contracting universe because “state-space volume is independent of physical volume.” The conclusion survives with a corrected argument. Entropy is a property of the state, and the available multiplicity $N(E)$ is set by the spectrum, not by the scale factor; but whether the state’s *actual* complexity continues to grow during contraction is a dynamical question, answered affirmatively only under Conjecture C4 (the cascade) plus an energy budget that does not collapse. The statement is conditional: if the cascade persists and the cutoff $E_{\max}$ does not shrink with the scale factor, the arrow persists through a Big Crunch, Gold’s reversal fails [4], and inhabitants of a putative reverse-arrow region would experience their own local multiplicity gradient as “the future”—the indexical resolution, which carries over once the cascade has an engine. If instead the cutoff is geometric (as *holographic* bounds suggest it might be, Conjecture C3), then contraction eventually forces decreasing multiplicity, and the framework itself predicts a Gold-like reversal. The two possibilities are separated cleanly by whether $E_{\max}(A)$ in (42) is a function of area alone: the conjecture form of the holographic bound thus *decides* the Big Crunch question, which a flat assertion cannot.

9.7 Superdeterminism and locality

Section 7 settles the Bell argument. The residue is short to state. The framework preserves statistical independence (no cosmic conspiracy), preserves no-signaling exactly (tensor structure, Theorem 7.1), and sacrifices only local realism of individual prime modes. Superdeterminism is rejected not because it is inconsistent (it is consistent) but because it is epistemically self-undermining and unnecessary: the arithmetic sector produces Bell correlations through superposed integers at no cost to locality of signaling. The “adjacent nodes in a non-spatial network” metaphor is retained as interpretation (C12): spacetime emergence from entanglement structure would indeed render “where” questions ill-posed at the fundamental level, but this remains a conjectural picture of emergence, kin to causal set theory’s own program [41] and to the causal theory of time it must surpass [40].

10 The Consolidated Conjecture Register

The register consolidates every ambitious claim that lacks proof: numbered, graded, with a falsifier or an explicit interpretive exemption. Grades: **A** (mathematically precise; partial evidence exists; actively testable), **B** (well-formed; speculative; no current evidence beyond structural fit), **C** (interpretive; tethered to A/B entries; not empirically separable at present). The *Depends on*

column lists register entries whose failure collapses the entry; the qualifier “(int.)” marks a dependency that is interpretive rather than logical—such an entry survives the parent’s failure as an autonomous mathematical problem. The falsifier column is typed by how the test is decided: **M** (mathematical: decidable by proof or certified finite computation, noiseless and repeatable), **P** (physical: decidable by comparison with measured data, statistical, one-shot), **M/P** (both routes exist); the classification is developed in Section 11.2.

ID	Statement	Grade	Depends on	Falsifier / test route
C1	<i>Spectral equivalence.</i> The renormalized near-singularity gravitational partition function of a 4-dimensional black hole equals ζ (and $\zeta_{\mathbb{Q}(i)} = \zeta L(\chi_{-4})$ in the Gaussian/5D sector), after conformal reduction to a self-adjoint H_{sing} with $H_{\text{sing}} \cong \hbar\omega_0 \log \hat{N}$.	A	— (root)	Derive H_{sing} from BKL/conformal reduction; compute its level statistics (cf. C5). Falsified if the near-singularity spectrum is provably not log-integer. Evidence base is thin—the structural Dedekind fit plus one reported program [3]; the grade reflects testability, not weight of evidence. M
C2	<i>Arithmetic cutoff.</i> The physically realized arithmetic universe is a finite or effectively finite truncation (primes $\leq X$, occupations $\leq K$), geometrically motivated, controlling the recurrence-time scale of Theorem 4.4.	B	C1	Fixes the return-time family: $\tau_\varepsilon \leq (2\pi C_F/\varepsilon)^{m_F}$ (Prop. 4.5), polynomial in the entropy S for states of fixed support. Falsified by recurrence phenomenology whose times grow faster than every power of the entropy at fixed resolution ε : enlarging the cutoff changes the bound only polynomially in S , so no cutoff-consistent enlargement absorbs an exponential-in- S recurrence scale. Polynomial-in- S returns at bounded support are the arithmetic prediction. M

ID	Statement	Grade	Depends on	Falsifier / test route
C3	<i>Holographic arithmetic bound.</i> $E_{\max}(A) = \frac{\hbar\omega_0}{4\ell_P^2} A + \hbar\omega_0 s(A),$ $s(A) = O(\log \frac{A}{\ell_P^2})$ universal (Conj. 6.1); with Theorem 6.1 it yields $S_{\text{BH}} = k_B(A/4\ell_P^2 + s(A)) +$ $O(k_B e^{-E/\hbar\omega_0})$ (Thm 6.2).	A	C1	Positive evidence: the proved conditional The- orem 6.2 (S_{BH} follows exactly once the bound is granted)—the grade certifies that conditional, not independent support. The counting defect is exponentially small in E (Thm 6.1(iii)), so every $\log(A)$ -class correction must be carried by the dictionary term $s(A)$; the pure model ($s = O(1)$) pre- dicts none. Mismatch with computed corrections (e.g. known $-\frac{3}{2} \log A$ terms) falsifies the pure form and measures $s(A)$. M

ID	Statement	Grade	Depends on	Falsifier / test route
C4	<i>Cascade/ETH, interacting completion.</i> The <i>interacting completion</i> $H_{\text{tot}}^{(U,V)} = H_P + H_W + H_{\text{hop}} + U \sum_{p<q} \hat{n}_p \hat{n}_q + V \sum_{p<q} (\hat{n}_p + \hat{n}_q)(a_p^\dagger a_q + a_q^\dagger a_p)$ (diagonal or kinetic) thermalizes on the prime lattice: $\sigma_{\text{ETH}}(D) \rightarrow 0$ and diagonal $\rightarrow$ microcanonical along truncation sequences at fixed dose (Conj. 3.1). The bare H_{tot} is excluded (Remark 3.12).	A	C1 (int.)	<i>Scaling criterion:</i> ETH diagnostics along truncation sequences with $D \rightarrow \infty$ at fixed effective dose VK^2 — sparse eigen-windows to $D = 5.9 \times 10^4$, a factorization-free Chebyshev tier past the LU fill ceiling to $D = 3.8 \times 10^4$ in $d = 4$, an upper-edge Krylov window to $D = 1.05 \times 10^5$ (Remark 3.12(j)), and Krylov equilibration to $D \approx 10^5$; falsified if the eigenstate fluctuation never approaches the thermal scaling and $\text{diag} - \text{micro}$ stays $O(1)$ for every interacting completion. Current evidence: the bare system is decided—structurally non-ergodic at all scales computed; the <i>diagonal</i> quartic localizes monotonically ($\text{PR}/D \rightarrow 10^{-3}$); the <i>kinetic</i> completion is the only one that relaxes (occupation falls from K to ≈ 3.5 within $\tau \approx 20$, GOE statistics, delocalized eigenstates at matched dose), and its finite-time plateau equals the diagonal ensemble exactly where the latter is computable ($D \leq 10^4$, $\text{plateau} - \text{diag} /K \leq 0.005$); but neither scaling leg closes: in absolute units the eigenstate fluctuation decays as $\kappa \propto D^{-0.185 \pm 0.021}$ at $d = 3$ and $D^{-0.22} \rightarrow D^{-0.26}$ at $d = 4$ (four points to $D = 3.8 \times 10^4$, the last past the LU ceiling) against the strong-ETH $D^{-1/2}$, its distance from the fixed-fraction thermal benchmark grows ($4.6 \rightarrow 11.3$ at $d = 3$, $4.6 \rightarrow 7.8$ at $d = 4$; Remark 3.12(h)–(i)), the

ID	Statement	Grade	Depends on	Falsifier / test route
C5	<i>Arithmetic–chaos correspondence.</i> The spectral statistics of the quantized near-singularity (BKL-type) Hamiltonian match GUE, and its spectral determinant is arithmetic: the Hilbert–Pólya operator for the zeta zeros [43–45].	B	C1 (int.)	Compute pair correlations of quantized cosmological billiards; a proven non-GUE spectrum for the singularity sector falsifies; conversely, any rigorous Hilbert–Pólya operator would confirm the bridge (and RH’s spectral side). On the lattice model itself: GOE ratio statistics at generic coupling for $D = 625$ to $D = 5.9 \times 10^4$ (eigen-windows; equilibration diagnostics to $D \approx 10^5$), with a Poisson control at weak coupling and label-scrambled controls at fixed density of states (Remark 3.12(f): matched-dose statistics scramble-insensitive, generic-family $\langle r \rangle$ and PR/D arithmetic-sensitive). The finite-size trend of $\langle r \rangle - r_{\text{GOE}}$ in $1/\log D$ does not settle the limiting value (Remark 3.12(g)). M
C6	<i>Hagedorn wall as singularity resolution.</i> Approach to a spacetime singularity is approach to T_H : energy converts to primon creation, curvature growth saturates (Remark 6.5).	B	C1, C3	Near-singularity GR calculations (e.g. BKL maps) exhibiting equilibration at a critical scale; falsified by derived continua with no critical behavior. M
C7	<i>Janus boundary condition.</i> The universal state’s history touches the unique zero-entropy floor $ 1\rangle$ at one end (the arithmetic Past Hypothesis).	B	C1, C4	No direct route; constrained by C1, C4; cosmological tests indirect (initial-state smoothness statistics). P
C8	<i>Two-sided entropy.</i> The entropy profile is U-shaped about $ 1\rangle$ (Janus-point cosmology, arithmetic form) [37, 38].	B	C7	As for C7; additionally, CMB statistics distinguishing one-sided from two-sided low-entropy origins. P

ID	Statement	Grade	Depends on	Falsifier / test route
C9	<i>Galois–gauge dictionary.</i> Minimal form of Conj. 8.1: charge lattice $\leftrightarrow$ splitting restrictions; branching ratios $\leftrightarrow$ Chebotarev densities; hierarchy $\leftrightarrow$ ramification.	B	C1 (int.)	<i>Calibrated dictionary protocol</i> (restated): fit all channel ratios of ≥ 5 independent decay channels at precision $\sigma \leq 0.3\%$ with one common group order $N = G \lesssim \text{few} \times 10^2$ and class sizes $c_i = C_i $ (each $ C_i $ divides $ G $ by orbit–stabilizer, so the Chebotarev densities are exactly c_i/N); Monte-Carlo-calibrated false-positive rate $< 1\%$ on random tables. The protocol is well-formed only when the channel set, the class-to-channel identification, and the group-order range are fixed before the data are examined: the calibration prices the fit, not the analyst’s freedom. The frozen protocol has been applied to the Z -pole five-channel table: $T = 4.1 \times 10^5$ at the null’s 59th percentile—signature absent at the Z scale with detection power 1.000 over the whole registered range at current precision (each channel excludes individually at $\geq 2.8\sigma$), an exclusion rather than an under-powered null, compatible with the conjecture’s unification-scale placement and carrying no confirmatory content. A seven-channel granularity variant has been executed under the same discipline: a frozen, committed 2024 PDG extract (seven-flavour Z -width decomposition; null calibration run before the fit; empirical false-positive rate 0/3000 at the $\chi^2_{0.95}(7)$ threshold; injection power demonstrated at 3×10^5 events with the divisor-scan’s highly-composite- N

ID	Statement	Grade	Depends on	Falsifier / test route
C10	<i>CP violation as the σ-odd sector.</i> CKM phase and baryon asymmetry are the arithmetic index of asymmetric couplings $w(\pi) \neq w(\bar{\pi})$ (Prop. 8.5); the ratchet compounds multiplicatively along the cascade [36].	B	C4	Structural constraint (one-sided, necessary only): any derived CP/T observable must factor through split/inert statistics; a derivation forcing a genuinely non-arithmetic phase breaks it. No CP observable is derived at present; the entry is tested only through C4’s dynamics (the ratchet lives in the non-conserving H_W sector, Remark 3.6) and through the arithmetic-factorization requirement. M
C11	<i>Arithmetic hierarchy.</i> $\log(M_{\text{Pl}}/M_{\text{EW}}) \asymp E_{\text{cut}}/2\hbar\omega_0$, i.e. $E_{\text{cut}} \asymp 74\hbar\omega_0$ (Conj. 6.2): the hierarchy is a cutoff question.	B	C1, C3	Any derivation fixing ω_0 to a known scale (e.g. a horizon temperature, C3) closes or breaks the numerology; conversely a derived 10^{32} -state effective arithmetic universe confirms. M
C12	<i>Perspectival Everett semantics.</i> Branches are factorization histories in $\mathcal{H}_{\mathcal{A}}$; “now” and “I” are indexicals; the felt flow is record formation along the cascade [30, 42].	C	C4	Interpretive: tethered to C4; not separately falsifiable at present; internal criteria of coherence and derivative phenomenology only.

Three structural observations about the register. First, its *physical depth ordering* is $\text{C1} \Rightarrow \{\text{C2}, \text{C3}, \text{C5}, \text{C6}\} \Rightarrow \{\text{C4}, \text{C7}, \text{C8}\} \Rightarrow \{\text{C9}, \text{C10}, \text{C11}\} \Rightarrow \text{C12}$, with two cross-layer edges ($\text{C11} \leftarrow \text{C3}$: the hierarchy falsifier invokes C3’s horizon temperature; $\text{C10} \leftarrow \text{C4}$: the ratchet compounds along the cascade): the zeta/black-hole identification is the root, and its confirmation or collapse propagates deterministically through the physical plane. But the register is *two-plane*: on the mathematical plane, C4, C5, and the computational content of C9 (find a number field matching a branching table) are autonomous problems that survive any verdict on C1—which is precisely why they carry the register’s most economical tests. C4’s listed dependency on C1 is interpretive, not logical: the arithmetic sector and $H_{\text{tot}}^{(U,V)}$ exist unconditionally, so the thermalization problem is well-posed even if the spectral-identification program collapses. Second, the falsifier column is typed (Section 11.2): eight of the twelve entries (C1–C6, C10, C11) can be advanced or refuted without any experiment—by proof, or by finite computation with certified error—while the physical surface is concentrated in C9’s dated tests and the cosmology entries C7–C8. Third, the register is conservative: no claim appears both as a theorem and as a graded conjecture; where a strong reading is available but unproved, only the graded conjecture is carried.

11 Falsifiability, Predictions, and Test Programs

11.1 What the framework predicts that rivals do not

1. *No proton decay, by construction.* The arithmetic GUT unifies through number fields, not through a larger Lie group; baryon number is not gauged into a decay channel. Absence of proton decay, a persistent embarrassment for SU(5)-type models, is here a structural feature. Conversely, observed proton decay would falsify C9 outright—a clean, dated test.
2. *Branching statistics are group statistics.* Under C9, unification-scale branching data must organize into Chebotarev densities: rational numbers with denominators dividing a Galois group order. This is a sharp, unusual, and cheap-to-check signature. The check is pre-registered and calibrated: a class-equation-constrained divisor fit over group orders $N \leq 360$, with a Monte-Carlo-calibrated null at the measured uncertainty pattern (the divisor constraint is what saves the test; the weak form—each ratio near some rational—is vacuous because $\mathbb{Q}$ is dense) and with decision rule and power tables fixed in advance. Applied to the Z -pole five-channel table, the signature is absent with detection power 1.000 over the registered range at current precision: the exclusion holds at the Z scale, which is compatible with the conjecture’s unification-scale placement but carries no confirmatory content. The instrument is validated and its power is measured; the purchase waits on unification-scale data, and the frozen protocol applies to future tables. At seven-channel granularity (a frozen, committed 2024 PDG Z extract; null calibration before the fit; empirical false-positive rate 0/3000 at the $\chi^2_{0.95}(7)$ threshold) the verdict is unchanged: no common- N table fits, and the precision tables without modification (a Z factory improves the statistical floor by $\sim 3 \times 10^{-3}$). The protocol is well-formed only when the channel set and the group-order range are fixed before the data are examined: the Monte-Carlo calibration prices the fit on random tables, not the analyst’s freedom in choosing channels, identifications, or N . Two 2024 PDG tables beyond the Z pole are now on record under the same discipline (frozen extracts committed before their fits): the W -boson four-channel table excludes—no common- N table fits its fractions at the 1–2% precision, with the injection check at full power (100/100 recovery)—and the Higgs seven-channel table fires the decision rule ($\chi^2 = 0.333$, $p < 1/3000$) in a regime where its channel uncertainties are two orders of magnitude coarser than the calibrated one and its values are Standard-Model-coupled evaluations: the anomaly is recorded as flagged, not confirmed, with the recovery power 0/100 and a post-hoc bootstrap diagnostic (3.5%) documented alongside it. Precision, not machinery, is what the dictionary now waits on.
3. *A single new scale.* The framework has exactly one dimensionful parameter, ω_0 , with $T_H = \hbar\omega_0/k_B$. Every other number in the physics layer is either an integer, a group order, or a value of an L -function. Any phenomenological use of the framework that requires a second free scale has already left the theory.
4. *The counting contributes no logarithmic correction.* The defect of Theorem 6.1 decays like $e^{-E/\hbar\omega_0}$: integer-lattice counting contributes no $-\alpha \log A$ term at all, so any correction of that class in the black-hole entropy must be carried entirely by the dictionary term $s(A)$ of Conjecture C3. The pure model ($s = O(1)$) predicts none; computed corrections of the known $-\frac{3}{2} \log A$ type would falsify the pure form and measure s (register C3’s falsifier). The fluctuation sector the counting *does* contribute is the prime-counting oscillation term, bounded by $O(\sqrt{x} \log^2 x)$ under RH (Section 6).
5. *Recurrence phenomenology is layered and cutoff-controlled.* The framework’s cosmology distinguishes exact recurrence (forbidden for generic states, Theorem 4.2) from ε -recurrence (universal for the free flow, Theorem 4.4, but with return times growing without bound in the state), and identifies genuine transience only in the cascading dynamics with three or more active modes (Proposition 4.6). Any completed theory of the arithmetic cutoff must

fix where our universe sits among these layers—a prediction in the sense that the cutoff mechanism cannot be indifferent.

11.2 Falsifier classes

The register’s falsifier column mixes two kinds of tests, and the difference is not cosmetic. A *mathematical* falsifier is decided by a proof or by a finite computation with certified error: it is noiseless, repeatable, and can be re-run by anyone; its only uncertainty is truncation, and the computation itself certifies the truncation (residual norms, certified bounds, exact integer arithmetic). A *physical* falsifier is decided by comparison with measured data: it carries measurement uncertainty, it is one-shot (Nature does not repeat the experiment), and it is exactly where analyst freedom can corrupt inference—which is why the physical tests of this framework are pre-registered (statistic, null, decision rule, and power fixed before the data are examined; the Z -pole application of C9 is the executed template). The classification:

Entry	Class	Decisive instance	Uncertainty
C1	M	derived H_{sing} spectrum not log-integer	proof
C2	M	recurrence growth vs. polynomial-in- S bound	truncation, quantified
C3	M	$s(A)$ vs. computed $\log A$ corrections	literature-exact
C4	M	ETH scaling along certified truncations	finite-size, residual-certified
C5	M	GUE pair correlations; Hilbert–Pólya construction	finite-size
C6	M	BKL-map analysis blocking critical behavior	model class
C7	P	initial-state smoothness statistics	cosmological
C8	P	CMB one- vs. two-sided low-entropy statistics	cosmological
C9	M/P	dictionary closure (M); proton decay, precision tables (P)	MC-calibrated / measurement
C10	M	derived CP observable not factoring through split/inert	model class
C11	M	derivation fixing ω_0 to a measured scale	measured input
C12	—	interpretive; internal criteria only	—

Two structural facts follow. First, the framework’s falsifier surface is predominantly mathematical: eight of the twelve entries are decidable without any experiment, which is unusual for a framework with cosmological ambitions and is a direct consequence of its arithmetical form—the physical claims compress into number-theoretic statements whose truth is a matter of proof or computation. Second, the physical surface is small, concentrated, and dated: proton decay (C9), unification-scale precision branching (C9), and the low-entropy statistics of the cosmic background (C7–C8). Where the two classes meet—C9’s dictionary applied to measured tables—the mathematical side supplies the calibrated instrument and the physical side supplies the data, and the discipline on both is pre-registration on the physical side and certified error on the mathematical side.

11.3 Mathematical test programs

Four problems, in increasing order of depth, would each move multiple register entries:

1. *Prove or refute thermalization for $H_{\text{tot}}^{(U,V)}$ on finite truncations (C4)*: a numerical program of ETH tests along the prime lattice, directly analogous to established kicked-rotor and Bose–Hubbard studies; every ingredient is finite-dimensional and computable. The program has been carried out at three scales and for both completions: dense diagonalization to $D = 4096$,

sparse shift-invert eigen-windows to $D = 5.9 \times 10^4$, and Krylov equilibration to $D \approx 10^5$ —all laptop-scale (minutes per configuration; the interior eigensolver tier is bounded by LU fill, the evolution tier only by $D\tau$; diagonal pivoting, needed for the kinetic family, cut the generic factorization from 257 s to 3 s with eigenpair residuals $\leq 6 \times 10^{-8}$). A factorization-free window tier—block Chebyshev subspace iteration on $(H - c)^2$ with c the median of the density of states, the passband sized by stochastic Lanczos quadrature, every selected pair certified by its residual—reproduces the LU windows to machine precision where both run ($\max |\Delta\lambda| \leq 4 \times 10^{-11}$ at $D = 10^4$, κ within 0.6% at $D = 6.9 \times 10^3$; Remark 3.12(i)); its cost ceiling is characterized by the Chebyshev degree $\ln(\varepsilon) R/(2t)$ at the dose-pinned radius R . Verdict: level statistics GOE at generic coupling throughout; the eigenstate fluctuation ratio $\sigma_{\text{ETH}}/\text{std}$ is flat in D (≈ 0.95 for the bare, quartic, and kinetic systems alike, at matched dose across $D = 3375 \rightarrow 59319$), and in absolute units the fluctuation decays as $D^{-0.185 \pm 0.021}$ ($d = 3$), two and a half times more slowly than the strong-ETH $D^{-1/2}$ (Remark 3.12(h)); the diagonal quartic completion localizes monotonically in U ; the kinetic completion (density-assisted hopping, Hermitian and number-conserving, self-adjoint in the infinite-volume limit by Proposition 3.9 and validated to 10^{-15} against a first-principles dense assembly) is the only one that relaxes—fast initial decay to ≈ 3.5 within $\tau \approx 20$ with GOE statistics and delocalized eigenstates, and a finite-time plateau that equals the diagonal ensemble exactly where the latter is computable ($D \leq 10^4$)—but the diagonal ensemble itself stays 0.04–0.09 of K from the microcanonical value at matched dose with no closing trend, and the $D = 20736$ trajectory extended to $\tau = 300$ saturates at $\langle \hat{n}_d \rangle_{\text{diag}} \approx 4.40$ against a microcanonical value of 3.35. In absolute units the eigenstate fluctuation decays as $D^{-0.185 \pm 0.021}$ at $d = 3$ ($D^{-0.222}$ at $d = 4$) against the strong-ETH $D^{-1/2}$, with the distance to the fixed-fraction thermal benchmark growing by a factor ≈ 2.5 along the ladder and effective random-combination dimensions $N_{\text{eff}} \approx 8 \rightarrow 23$ versus shells of $\sim 10^3$ states (Remark 3.12(h)); the equilibration leg requires the $D \rightarrow \infty$ limit of the diagonal ensemble, which shows no convergence at accessible scales. The mathematical core of the question has since been reduced to the occupation sectors (Prop. 3.10): the number-conserving core is a direct sum of finite sector ladders, exactly block-tridiagonal in the layer grading, with a single interaction (the kinetic term) whose relative strength is a control parameter—decaying as $1/S$ at matched dose, dominant at fixed V . The two-mode sector is closed in closed form (an obstruction by solvability, with an analytic fluctuation floor); in $d = 3, 4$ the sector ladders and the classical limits of the sector problem on $\mathbb{C}P^{d-1}$ (integrable one-body flow at matched dose; chaotic kinetic flow at fixed V) frame the remaining question, and the joint thermalization signature is absent at every accessible sector dimension (Remark 3.12(l)).

2. *Compute the level statistics of the interacting arithmetic Hamiltonian and compare with GUE* (C5): the prime lattice with log-spaced on-site energies is a banded, incommensurate system; its statistics are accessible by dense-diagonalization experiments and by Dyson–Brownian-motion arguments. The banded-incommensurate claim is verified numerically at every scale computed: GOE ratio statistics ($\langle r \rangle = 0.45\text{--}0.58$) from $D = 625$ to 5.9×10^4 at generic coupling, with the weak-coupling limit identifiable by $\text{PR}/D \rightarrow 10^{-4}$ —though whether the correspondence is GOE or GUE in the infinite limit is exactly what finite windows cannot decide ($\langle r \rangle$ separates GOE from GUE by only 0.06).
3. *Construct a Hilbert–Pólya candidate from the arithmetic sector*: the primon Hamiltonian itself has spectrum $\log n$, not the zeros; the zeros enter as resonances of the *fluctuation* sector $-\zeta'/\zeta$. A candidate operator whose spectrum is $\{\frac{1}{2} + i\gamma\}$ built from the walk/hop sector would connect the framework to the deepest open problem in the mathematics (C5, and with it the spectral reading of RH via von Koch’s equivalence [20]).
4. *Close the Chebotarev dictionary on one completed instance* (C9): find a number field whose splitting statistics reproduce a measured branching table. Even one instance would convert

the dictionary from analogy to theory; the search space is computable and finite for any fixed table. The decision machinery—the constrained fit, the calibrated null, and the power tables—is in place and validated by injection; what is missing is data at the dictionary’s scale. The instrument now runs at two granularities: the five-channel scan and a seven-channel variant executed against a frozen, committed 2024 PDG extract (pre-registered null-first protocol; the seven-flavour Z -width decomposition; injection power at 3×10^5 events with the divisor scan’s highly-composite- N preference documented); both reject every common- N table at the Z scale, as the conjecture requires. The runner is table-agnostic: a new precision table is frozen with the embedded protocol, committed, and fitted in minutes; the W four-channel application (full injection power) rejects every common- N table at the W scale, and the Higgs seven-channel application flags a proximity anomaly at $\chi^2 = 0.333$ that the 8–50% channel uncertainties and their Standard-Model-coupled provenance leave uninterpretable—the instrument is ready for the precision that would decide it.

11.4 Physical test programs

The physical falsifiers are few, and each names its data source and its decision instrument.

1. *Proton decay* (C9, dated). Observation of any proton decay channel falsifies the arithmetic unification outright—baryon number is not gauged into a decay channel anywhere in the dictionary. The instrument is the existing large-volume water Cherenkov and scintillator programs; no framework-specific analysis is needed, which is what makes the test clean.
2. *Unification-scale precision branching* (C9, protocol frozen). The class-equation-constrained divisor fit, its Monte-Carlo-calibrated null, and its power tables are fixed and validated (Section 11, item 2); the protocol applies without modification to future five-channel-or-better tables at $\sigma \leq 0.3\%$. A Z factory improves the statistical floor by $\sim 3 \times 10^{-3}$; unification-scale decay channels would enter through the same fit. A positive identification requires the multi-channel closure (Section 11.3, item 4); a single channel can only exclude.
3. *Low-entropy statistics of the cosmic background* (C7, C8). One-sided versus two-sided low-entropy origins leave different statistics in the CMB temperature and polarization maps; the framework’s Janus prediction is two-sided. The instrument is standard observational cosmology; the framework contributes the prior, not the telescope.
4. *Scale identification* (C11, input side). Any derivation fixing ω_0 to a measured scale (a horizon temperature, a Hagedorn scale) closes the framework’s one free parameter or breaks the numerology; the measurement enters as the input, the derivation is the mathematical side.

Each physical test inherits its discipline from the class analysis of Section 11.2: the decision rule is fixed before the data, the uncertainty is quantified (Monte-Carlo null, power table), and the outcome is reported as exclusion, identification, or under-powered, never as confirmation by compatibility.

11.5 Assessment

The framework now satisfies three standards. Its mathematical core is genuinely theorem-grade: the spectrum, the partition function, the Hagedorn asymptotics, the counting law, the clock kernel, the no-periodicity theorem, the occupation-sector self-adjointness of the interacting completions (Proposition 3.9), and the entanglement structure are proved. Its analogies are tightened: every structural analogy is either a citable theorem (Kronecker–Weber, Chebotarev, Erdős–Kac, Pólya, Bohr) or an explicit register entry. And its remaining ambition—that the primes order the spectrum of spacetime itself—is stated in a form precise enough to be wrong: C1 through

C12 name their evidence, their dependencies, and their falsifiers, the falsifier column is typed (Section 11.2), and the two numerically testable entries (C4, C9) carry carried-out programs with stated uncertainties. A framework that can be wrong in stated ways is a research program; one that cannot is a metaphor.

A Notation

Symbol	Meaning
Pr	rational primes; $\mathcal{P}_{\mathbb{Z}[i]}$ Gaussian primes modulo units
$\mathcal{F}$	free commutative monoid on Pr (occupation vectors)
$\mathcal{H}_{\mathcal{A}} = \ell^2(\mathcal{F})$	arithmetic Hilbert space (Def. 2.1)
$ n\rangle$	occupation state of the integer $n = \prod p^{k_p}$
$\hat{n}_p, \hat{N}_{\text{tot}}, \hat{N}$	mode occupation; total primon number; multiplicative number operator
$H_P = \hbar\omega_0 \log \hat{N}$	primon Hamiltonian (Def. 2.3)
$T_H = \hbar\omega_0/k_B$	Hagedorn temperature
M_n	multiplication operator, $ k\rangle \mapsto nk\rangle$ (Def. 3.2)
H_W, H_{hop}	walk and hop Hamiltonians; the prime lattice (Sec. 3.3)
$H_{\text{kin}}(V)$	kinetic completion $V \sum_{p < q} (\hat{n}_p + \hat{n}_q)(a_p^\dagger a_q + a_q^\dagger a_p)$ (Conj. 3.1)
$\zeta, L(\chi_{-4}), \zeta_{\mathbb{Q}(i)}$	Riemann zeta; Dirichlet $L \pmod{4}$; Dedekind zeta of $\mathbb{Q}(i)$
$\omega(n), \Omega(n)$	number of distinct / total prime factors
$N(E) = \lfloor e^{E/\hbar\omega_0} \rfloor$	state counting function (Thm 6.1)
$ \tau; \beta\rangle, K(\Delta\tau)$	zeta clock family and two-time kernel (Def. 5.1, Prop. 5.2)
C, P, T	arithmetic charge conjugation, parity, time reversal (Def. 8.4)
C1–C12	the conjecture register (Sec. 10)

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